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REVIEW

Federated Learning in Multimodal Healthcare Diagnostics: Privacy-Preserving AI for Biomedical Imaging, Electronic Health Records, Wearables, and Clinical Decision Support

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Abstract
The rapid expansion of chemical and biomedical imaging-from Magnetic Resonance Imaging (MRI), Computed Tomography (CT), ultrasound, and histopathology to biosensor and wearable outputs-allows unparalleled opportunities for early diagnosis, precision monitoring, and personalized care. Traditional AI pipelines, however, rely on the centralization of sensitive clinical data, which in general is not feasible because privacy regulations, institutional policies, and variable hardware across imaging systems often prohibit it. Federated learning (FL) overcomes such limitations by enabling multi-institutional training of AI without transferring raw images or patient records. Instead, local models learn from site-specific imaging and physiological data, and only model updates are shared across collaborating sites, thereby reducing direct exposure of raw patient data and improving regulatory alignment; however, confidentiality is not automatic and depends on additional safeguards against risks such as gradient leakage, model inversion, and membership inference attacks. In this review, we conduct a literature review of FL in healthcare including practical applications from multiple areas of study, Biomedical Imaging, Electronic Health Records, Physiological/Chemical Signals, Chronic Disease Management, Elder Care, and Clinical Decision Support, among others. In addition to providing insight into the successes, we outline the key barriers to implementing FL including system heterogeneity, data imbalance, communication bottlenecks, and issues related to fairness. Tying together research and practice we also offer actionable advice to regulators, developers, and healthcare systems for how to move forwards towards safe, equitable and cross geographical FL systems in medicine.

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1 Introduction

AI/ML are the growing trends, and healthcare stands as one of the most developing industries because of these technologies [1], [2–4]. Large volumes of health information are generated each day. Examples include electronic health records (HER), diagnostic biomedical images like computed tomography and magnetic resonance imaging, data from wearable sensors, remote monitoring tools, etc [5–8]. All this information will make it possible for doctors to make quicker and wiser decisions, diagnose diseases more quickly, and come up with more appropriate treatment programs suitable for a particular patient [9–12]. Since health data is very sensitive and personal in nature, it creates challenges around privacy, ownership and secure sharing. Regulations like Health Insurance Portability and Accountability Act (HIPAA) in the US and the General Data Protection Regulation (GDPR) in Europe mean that data cannot be moved to any place and combined into one central place to train a traditional ML system. Thus, even with the data available, it becomes a challenge to use it for training an AI model [13–15].

This is where FL comes in. FL is a new type of ML that operates differently from traditional ML. Instead of collecting all the raw data in one central server, FL allows each hospital, device, or clinic to train the AI model with its local data. Only an update to the model, e.g., better weights or parameters, are sent to a central system (or peer to peer) that combines it all into a stronger model across the globe. The original data never leaves the device or hospital's system [16–20]. This makes it much safer and easier to comply with privacy rules, while still leveraging data from various locations. Many examples of FL are already being used in healthcare [21–23]. For instance, consider if hospitals in different countries train their AI models to spot tumours from scans without ever leaking pictures of patients. This preserves privacy, and means the model learns from a broader variety of imaging machines and patients. The visual below shows the transition from centralised ML to FL and, further along, to a privacy-enhancing, interpretable, accountable, and adaptable healthcare AI ecosystem (Fig. 1). You can see how the FL tackles the fundamental challenges of data-handling, legal compliance, scalability, generalisation, and trust against conventionally trained AI models [24].

FL is also used with EHRs to predict risks, for example for sepsis or heart failure, even if hospitals have different software and data formats. Wearable devices too has a big use-case. Smartwatches and fitness bands that monitor heart rate, sleep patterns and physical activity. FL is used to help them learn how to detect things like irregular heartbeats right there on the device, without sending raw data into the cloud [25–27]. It also energizes clinical decision support systems (CDSS) in which hospitals work together to create frameworks that can notify doctors of early warning signs, like a sudden infection, without disclosing confidential patient information [28–31]. Furthermore, FL also represents broader trends that are currently defining innovation in healthcare artificial intelligence. For instance, with the rise of devices/sensors at the "edge" (or outside of traditional hospital settings), there is an emerging need to develop models that can operate at the "edge." FL can address this need. FL is also consistent with Explainable AI, where it is important to not only make predictions, but to explain why these predictions were made. New research is combining FL with various explainability tools to help doctors understand and trust models. On the security side - FL is fusing with blockchain and trusted execution environments (TEEs) to make the entire system sufficiently secure and auditable. This necessary for trust in AI systems [32–34].

Beyond the traditional biomedical approaches, an increasing number of diagnostic platforms are now dependent on chemical imaging. Techniques such as hyperspectral imaging, Raman and infrared microscopy, fluorescence-based molecular probes, and nanotechnology-enabled contrast agents provide information that is molecularly specific rather than restricted to simple anatomy [35–38]. These systems produce high-dimensional spectral signatures and chemically encoded contrast maps that are difficult to centralize owing to their massive size and patient-specific profiles. FL is a technique that has huge potential in this area since it can be utilized for joint learning to improve molecular contrast, spectral reconstruction, signature extraction, and response quantification without the exchange of raw chemical images from one center to another. As such, FL enables not only privacy-preserving diagnostics but also informs the integration of chemical sensing with imaging-derived molecular information [39–42].

The tools for creating FL systems are also improving. Libraries such as TensorFlow Federated, PySyft, and Flower have been making it easier for researchers and companies to experiment and implement FL. The libraries support different network structures, handle device dropouts, and integrate with other technologies for encrypting data. More often, hospitals and companies are trialling FL as a more secure mechanism for using AI in healthcare FL resolves a lot of problems, but is not without its own challenges [43–45]. The first one is that of system heterogeneity - that is to say, different devices (or indeed different hospitals) may have different internet access speeds and/or different computer capabilities. The second one is that of data heterogeneity - that is to say, the data that comes back from the devices may simply not have the same distribution. That may have a big impact on the way that the model is being trained. The communication of all of these devices takes time and

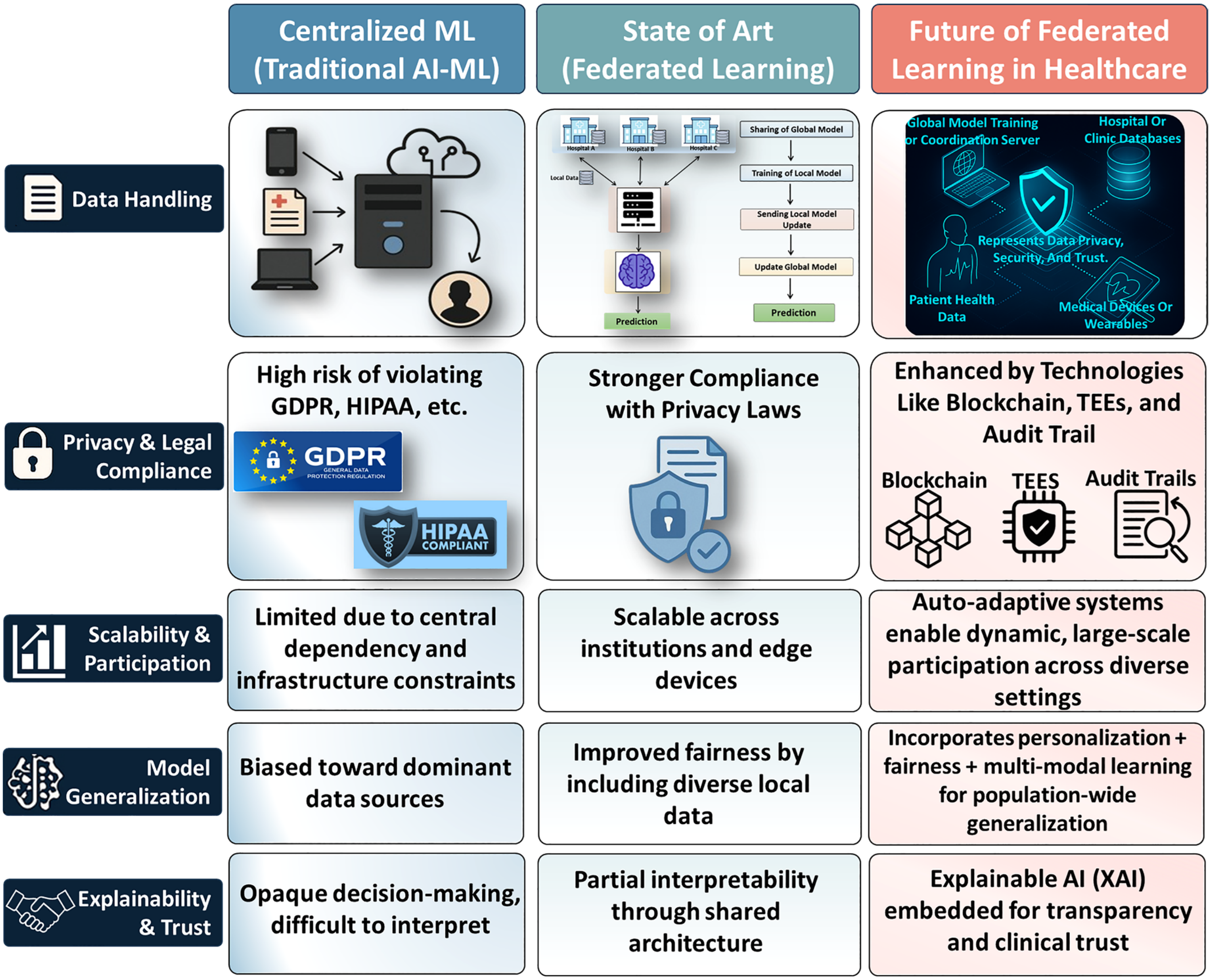


Fig. 1 Comparative overview of centralized ML, current state-of-the-art FL, and future directions in healthcare

bandwidth, and that slows everything down. Finally, there is the matter of fairness - that is to say, making sure that the models do not end up favoring some patient groups over others [46–48]. Looking ahead, FL has many exciting directions. One approach is to combine it with Large Language Models (LLMs) and multimodal data (such as text, images, and signals) to enhance the understanding of complex cases [49–51]. Another area of research in this field is self-adapting FL systems that can adjust their setting and parameters in response to the data, device, or user provided. As the field of FL is evolving, there is a need to establish standards and guidelines for FL systems to become legally operable in the health field. This review is aimed at providing a clear view of FL in the health field of AI.

Several previous reviews have considered certain facets of FL in healthcare, but they continue to be predominantly focused either on privacy-preserving solutions, algorithmic considerations, architecture designs, or particular types of clinical data. For instance, while focusing on privacy enhancement techniques, such as differential privacy and homomorphic encryption, Gu et al. have reviewed the use of FL in healthcare [52]; in their turn, Li et al. have highlighted methodological and privacy challenges associated with FL adoption in the medical sphere [53]. Li et al., in another study, provided a classification of FL systems from a viewpoint of systems engineering [54]. The works by Wen et al. and Li et al. discussed FL challenges and applications in regard to big data and structure medical data, respectively [55, 56]. Unlike previous studies, however, the current review gives consideration to a much more translatable approach by considering the use of FL in biomedical imaging, electronic health records, wearable devices, physiological and chemical signals, chronic disease management, and clinical decision-support systems under one framework of multimodal healthcare diagnostics. What makes this review novel is the applied orientation, FL performance comparison for various domains in healthcare, discussion of deployment tools and privacy-preserving techniques, indication of technology readiness levels, and tailored strategies for relevant stakeholders. Hence, the present review offers a more pragmatic approach to FL by considering FL not just as an algorithmic and privacy-preserving solution but rather a framework for building clinically viable and trustworthy AI systems. The rest of this paper is structured as follows: Sect. 2 states the basics of FL and we demonstrate its examples in use cases in Sect. 3. Section 4 tackles the challenges and possible remedies, and Sect. 5 concludes this paper and details future work.

2 Methods

2.1 Review Design and Rationale

This review was conducted following the framework of a scoping review that involved an organized literature search and qualitative synthesis of findings. Scoping methodology was chosen due to the large, diverse, and rapidly changing nature of the field of federated learning in medicine, which includes biomedical images, electronic health records, sensor-based wearables, physiological measurements, clinical decision support, and chronic disease monitoring. There was no intention of conducting a meta-analysis or evaluating the magnitude of effects in different pooled comparisons in the scope of this work.

2.2 Search Strategy

A structured literature search of the literature was performed on four major scientific databases: PubMed, Scopus, IEEE Xplore, and Web of Science, for literature published between January 2018 and November 2024. FL papers are rare before 2018, and thus the older papers were dropped. The search strings combined controlled vocabulary (e.g. MeSH terms) and free-text keywords. The query was the following: (federated learning OR decentralized learning OR distributed training) AND (healthcare OR medicine OR clinical OR medical imaging OR EHR OR wearable OR biosensor) AND (machine learning OR deep learning). Additional modality-specific terms (e.g., “MRI”, “CT”, “ultrasound”, “ECG”, “EEG”, “CDSS”, “chronic disease”, “IoMT”) were incorporated to ensure high recall across subdomains. Backward and forward citation chaining was performed to identify influential works omitted by the automated search.

2.3 Screening Process

All retrieved records were subjected to a three-stage screening process. In the first step, duplicates across databases were removed through automated and item-by-item manual checking of titles, DOIs, and author lists. In the second stage, two independent reviewers screened titles and abstracts to determine relevance to FL in healthcare; disagreements were resolved through discussion. Third, the full-text screening ensured that the selected studies met the inclusion criteria, which were: the study had to be an implementation or evaluation of FL technique in a health-related environment, use biomedical or clinically relevant data (real or simulated), written in English, and had to include results. Studies were excluded if they discussed FL only conceptually, addressed only blockchain or networking infrastructure without a healthcare task, presented generic FL algorithms without clinical application, or consisted solely of abstracts, theses, or non-peer-reviewed sources.

2.4 Inclusion and Exclusion Criteria

We included studies that use FL to address health-related issues such as medical images, physiological signal processing, risk prediction based on EHRs, wearable sensors, chronic disease management, elderly care, decision support systems, etc. Additionally, studies included in this scoping review had to present a quantitative evaluation of FL application. Excluded studies were those lacking a concrete FL implementation, those without biomedical relevance, those focused purely on cryptography or systems design without a clinical task, or those not providing empirical results. Non-English works, grey literature, and partially reported abstracts were similarly excluded. These criteria ensured that all included works contributed directly to understanding the methodological landscape of FL in healthcare.

2.5 Data Extraction and Structuring

For each eligible study, we extracted data using a structured template comprising bibliographic information, application domain, dataset characteristics, FL paradigm (horizontal, vertical, cross-device, cross-silo, transferbased), aggregation and optimization strategies, model architecture, characteristics of clients, handling of heterogeneity, privacy-enhancing techniques, evaluated metrics and limitations acknowledged. Two reviewers performed data extraction independently, resolving disagreements through consensus. Due to the heterogeneity in study design, dataset size, evaluation criteria, and goals of the approaches, we instead carried out a narrative synthesis rather than a meta-analysis. We group the studies that we have surveyed into clinically-relevant buckets: medical imaging, physiological monitoring, EHR-based prediction, wearable devices and IoMT systems, chronic disease and elderly-care applications, and clinical decision-support tools, and compare the studies within each group with respect to method novelty, approaches towards non-IID data, privacy guarantees, baseline comparisons, and metrics of real-world readiness. This synthesis provided the basis for identifying common challenges and mapping out priorities for follow-up research. A PRISMA-style flowchart was added to summarize the study selection process (see Fig. 2F), including database identification, duplicate removal, title/abstract screening, full-text eligibility assessment, reasons for exclusion, and final inclusion in the narrative synthesis. The studies presented in Tables have been chosen to be representative of the final list of eligible studies because of their relevance to health care, clear use of FL, quantifiable outcomes, privacy or aggregation procedures, scope of domains, and translational applicability.

3 Fundamentals of FL

Before delving into its medical applications, it's helpful to discuss the general principles and workflow of FL. This section describes the principles behind FL, explaining how the general mechanics of this method differ from standard machine learning, how decentralized training protects the users' data while maintaining model accuracy, and how breaking it down into its three components - local training, parameter aggregation, and global updating - helps us understand how FL puts hospitals, devices, and researchers in touch with one another in a secure way. These principles form the backbone for the subsequent subsections that detail how FL is practically applied across medical imaging, electronic health records, physiological monitoring, and decision-support systems.

3.1 Basic Workflow: Local Training, Parameter Aggregation, Global Updates

Compared to traditional AI/ML approaches, FL has some clear advantages when it comes to healthcare. In the traditional workflow, all data had to be brought to one place, cleaned, and model trained on that dataset. This is effective, but fraught with risks for healthcare, as data may be protected by privacy laws or institutional policies. Centralised data may not also represent some patient cohorts equally, as depicted in Fig. 2(A) [51, 57–61]. FL flips this idea on its head. Instead of forcing the data to move, FL lets the model do the moving. FL lets the model learn from whatever source it's got (urban hospitals, rural clinics, personal devices), but it keeps sensitive information on the down-low. FL ends up building more accurate and fair models, which work better on real-world healthcare problems (Fig. 2(B)). FL is the standardized method where ML models are trained on local data on a number of devices or servers without needing to exchange the data itself.

This fundamental workflow is based on three steps: local training, parameter aggregation, and global model updating [62–65]. These steps let data be shared between partners who are learning together, but not actually exchanged amongst each other, hence FL works well in areas where it is concerned health care. In the first step, each participant- a hospital, clinic, or a medical device- is using a common, shared (and generally pre-defined) ML model. Instead of uploading data from their patient's records to some central location, the technique transmits the model to each of the local sites (known as edge devices), which are then trained in parallel using only that institution's private data.

For example, a hospital may take the ECG data results carried on patients' wearables and use them locally to train its ML model how to detect early problems in cardiac arrhythmias. The raw ECG signals themselves never leave the hospital, keeping it inside such privacy laws as HIPAA in the USA and GDPR in Europe [66–68]. Once local training is done, each site sends back only the updated model parameters (the specific weights or gradients it learned), back to a central aggregation server. While the secret sauce of the local data is baked into those model parameters, no patient information is carried back. This step of aggregation of parameters is vital and is the way in which it is possible to learn from data that is distributed across different sites while at the same time taking care of the inherent data ownership and privacy issues. For instance, one hospital might contribute to improvements in the detection of heartbeat anomalies in elderly patients, and another hospital might improve the performance on arrhythmia detection in younger patients or those with co-morbidities [67, 69].

This information is gathered by the server from all the participants and the server calculates the average of all the information gathered. This new version of the global model contains all the information gathered from all the different sites. The server sends the new version of the global model back to the different sites, and the sites are now ready for the next round of training. With time, the accuracy of the model improves, making it more robust and able to make reliable predictions for different types of patients. This communication workflow enables FL to debug and create powerful AI systems that benefit from global collaboration while respecting institutional and individual boundaries. In practice, this workflow lets rural clinics, large urban hospitals, and even consumer health products contribute to and benefit from a single model without ever having to share sensitive patient data. That way, federated learning strengthens data security,

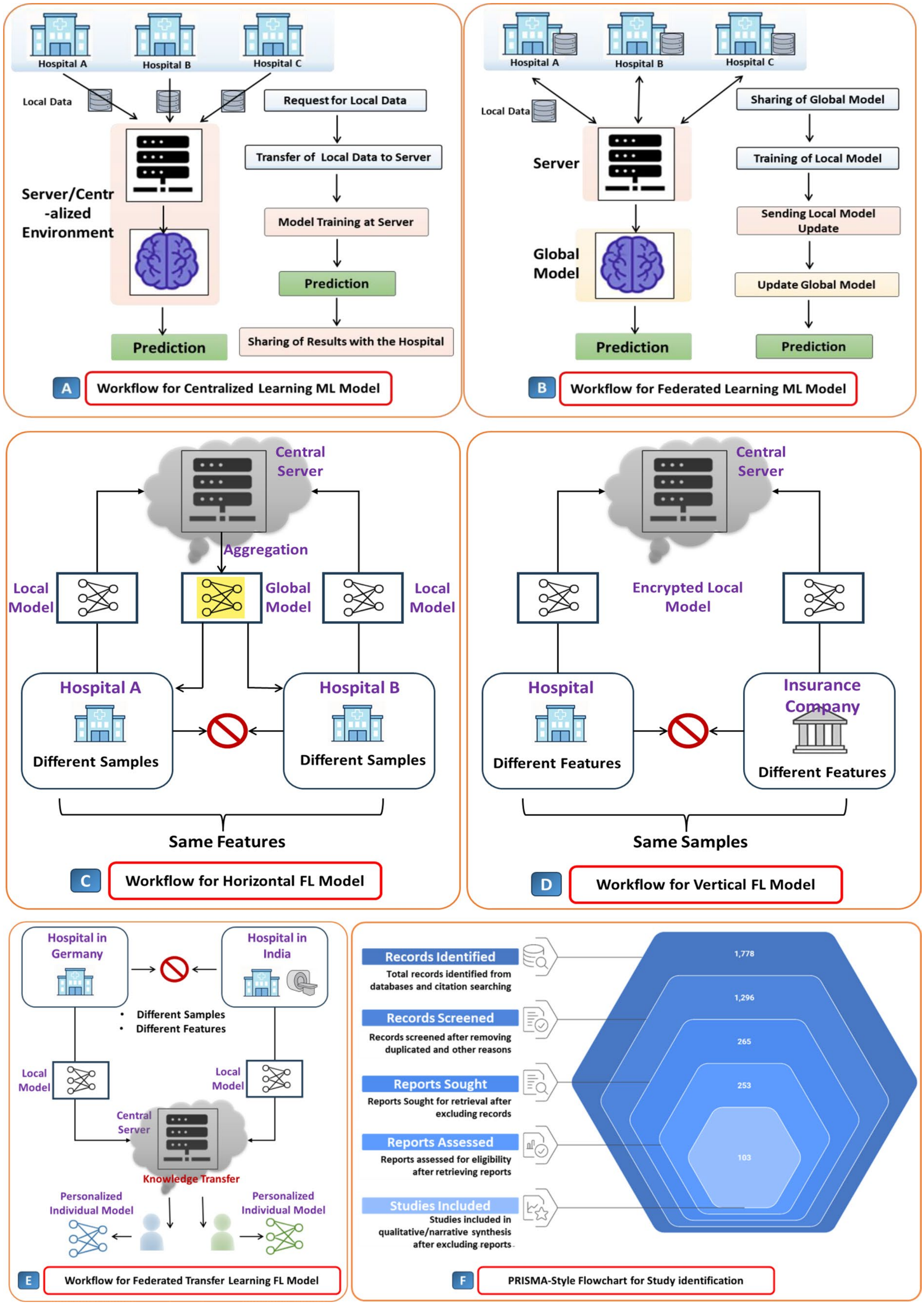

Hospital A
Hospital B
Hospital C
Local Data
Request for Local Data
Transfer of Local Data to Server
Server/Centr
-alized
Environment
Model Training at Server
Prediction
Prediction
Sharing of Results with the Hospital
A
Workflow for Centralized Learning ML Model
Hospital A
Hospital B
Hospital C
Local Data
Sharing of Global Model
Server
Training of Local Model
Sending Local Model
Update
Global
Model
Update Global Model
Prediction
Prediction
B
Workflow for Federated Learning ML Model
Central
Server
Aggregation
Local
Model
Global
Model
Local
Model
Hospital A
Hospital B
Different Samples
Different Samples
Same Features
C
Workflow for Horizontal FL Model
Central
Server
Encrypted Local
Model
Hospital
Insurance
Company
Different Features
Different Features
Same Samples
D
Workflow for Vertical FL Model
Hospital in
Germany
Hospital in
India
Different Samples
Different Features
Local
Model
Local
Model
Central
Server
Knowledge Transfer
Personalized
Individual Model
Personalized
Individual Model
E
Workflow for Federated Transfer Learning FL Model
Records Identified
Total records identified from
databases and citation searching
Records Screened
Records screened after removing
duplicated and other reasons
Reports Sought
Reports Sought for retrieval after
excluding records
Reports Assessed
Reports assessed for eligibility
after retrieving reports
Studies Included
Studies included in
qualitative/narrative synthesis
after excluding reports
1,778
1,296
265
253
103
F
PRISMA-Style Flowchart for Study identification

Fig. 2 Comparison between (**A**) centralized machine learning and (**B**) federated learning in a multi-hospital healthcare setting. (**C**) horizontal FL: same features, different patients across hospitals with model aggregation at the central server. (**D**) vertical FL: different features, same patients across hospitals and insurers, with an encrypted model collaboration. (**E**) Federated transfer learning: different features and patients with cross-domain knowledge transfer and personalization. (**F**) PRISMA-style flowchart of study identification, screening, eligibility assessment, and inclusion for this scoping review

regulatory compliance, and more inclusive and equitable healthcare AI models [70].

4 Applications of FL in Healthcare

Among the most data dense areas in medicine, imaging is a natural fit for federated learning. Many hospitals and research facilities have hundreds of thousands of scans across modalities, but sharing such data for centralised training is usually logistically impossible, as well as violating privacy regulations. FL makes it possible to share the value of these distributed datasets without risking breaching patient confidentiality. FL has been applied across multiple imaging methodologies from MRI and CT to ultrasound and histopathology, each with its own set of challenges and opportunities. The following sections highlight how FL is advancing these modalities, beginning with MRI as a leading case for privacy-preserving collaboration in robust diagnosis.

4.1 Different Medical Imaging Modalities

Medical imaging provides an excellent proving ground for FL since the datasets are also large, sensitive and, in general, heterogeneous across scanners, protocols and institutions. In the next few subsections, we look at examples of how FL is able to adapt to the demands of each specific modality, from MRI and CT down to ultrasound and digital pathology, while preserving maximum privacy and increasing generalization. We introduce notable architectures, aggregation schemes and personalization methods to medicine that mitigate non-IID distributions and device variance. Together, these modality-focused studies illustrate how the same FL fundamentals translate into reliable, deployment-oriented workflows across diverse clinical imaging tasks.

4.1.1 FL in MRI: Privacy-Preserving Collaboration for Robust Diagnosis

FL augments MRI diagnosis by allowing hospitals to co-train models on diverse, multi-site data without transferring raw images off-premises. This diversity aids models in learning domain-invariant features across scanners, protocols, and populations, which decreases brittleness while also cutting false positives/negatives [37]. FL also accommodates personalization; clients adapt the global model to local quirks, and carry out tasks such as missing-contrast synthesis and class-imbalance handling, which avert performance collapses when data are not IID. As only parameters and not patient data traverse the edge, FL enhances privacy and eases compliance while still-learning class-specific parameters from rare, distributed pathologies in different centers [71]. For patients, the benefits include earlier and more accurate detection, fewer repeat scans, improved consistency of interpretations across hospitals, and speedier updates to models based on changing evidence, all while ensuring privacy. In other words, FL transforms MRI networks to learning health systems that increase diagnostic quality while upholding patient trust [72, 73].

Recent works have shown that FL can achieve good performance on various MRI and imaging tasks with privacy guarantee. To illustrate this point, for instance, for the task of classification of brain tumor MRIs, the FL approach proposed in the work of Albalawi et al. adopts transfer learning in FL and hence attains high accuracy in the task while bypassing legal and IT issues on-site. Second, FL is also more robust against out-of-domain data, as shown in Fig. 3A. [58]. Extending beyond classification, Dalmaz et al. proposed a personalized FL framework for synthesizing missing MRI contrasts, enabling site-specific adaptation and improving cross-center generalizability under non-IID conditions (Fig. 3B) [75]. This is in contrast to the standard central model, as this method allows for per-site adaptation without the need to re-access the raw data. Given its design, which allows for customization at the site, it can handle scenarios with non-IID scanners and protocols. Liu et al. used vertebral segmentation in spinal imaging to demonstrate the application of the dual-attention federated learning method, which enhances the performance of multi-center datasets, especially in the presence of rare anatomy, without the need to share the raw data (Fig. 3C) [76]. FL has also been applied to functional MRI, where Li et al. demonstrated improved autism detection performance across sites by mitigating scanner and protocol shifts without centralizing data [37]. For oncologic prediction, FL-enabled multimodal fusion improved lymph-node metastasis assessment and allowed new centers to join collaboratively without complex data alignment (Fig. 3E) [78]. From prediction to secure thoracic CT detection, Heidari et al. trained blockchain-secured FL for lung-cancer detection on CT, strengthening auditability and privacy across sites [79]. This provides immutable compliance trails and lowers breach risk versus centralized repositories, while enabling near-real-time, cross-site updates. Finally, in breast imaging, memory-aware curriculum learning and transfer learning within FL improved

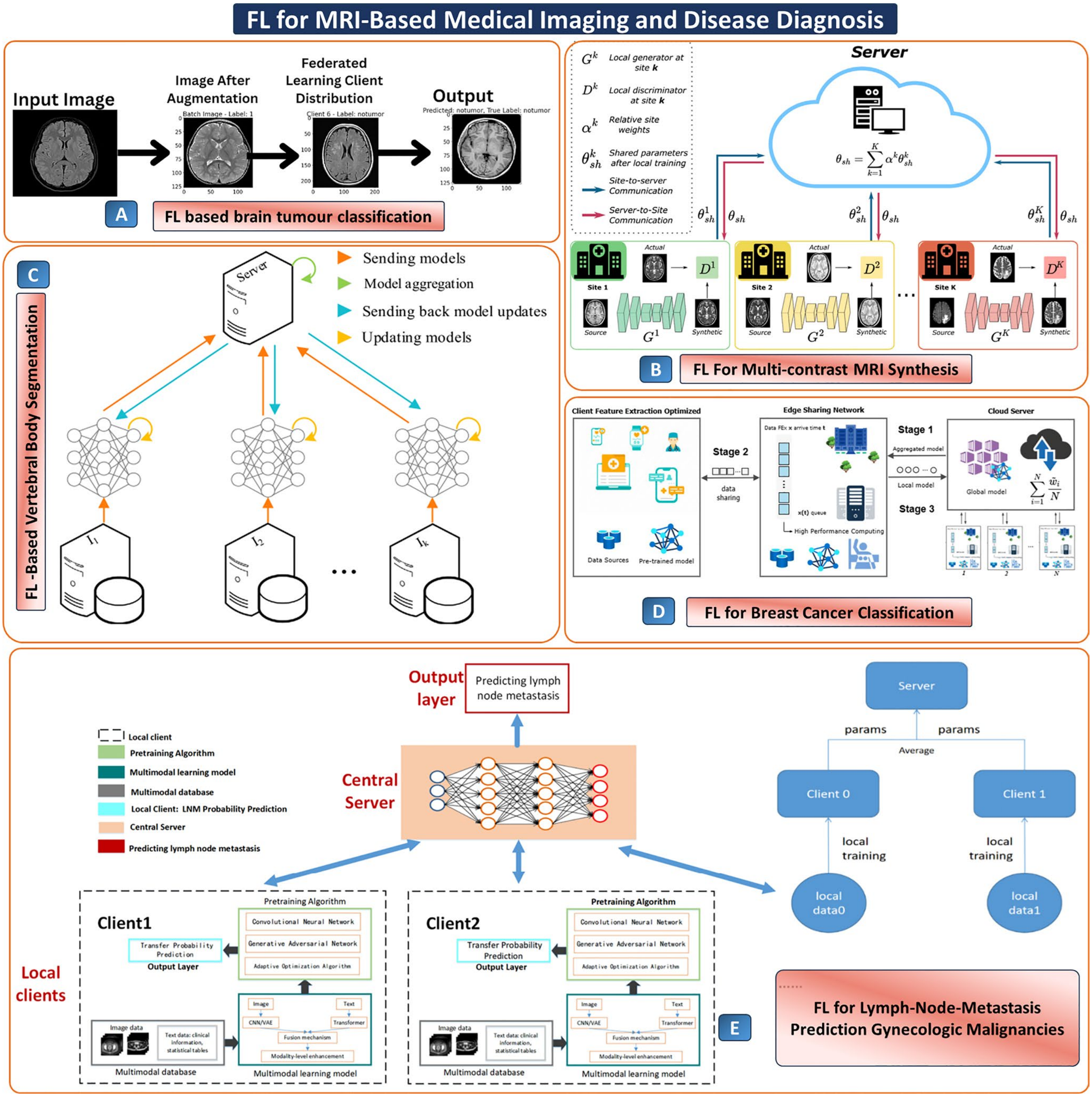


Fig. 3 (**A**) FL workflow for brain tumor classification using augmented MRI scans, taken from [74], with the permission of Springer. (**B**) Federated GAN-based architecture enabling multi-institutional collaboration for generating synthetic MRI contrasts from source images, taken from [75], with the permission of Elsevier. (**C**) Federated segmentation pipeline for spinal imaging, taken from [76], with the permission of Elsevier. (**D**) edge-cloud FL architecture for breast cancer classification, taken from [77], with the permission of IEEE. (**E**) a multimodal FL framework integrates imaging and clinical data across clients to predict lymph node metastasis using pre-trained models and a centralized aggregation mechanism, taken from [78], with the permission of MDPI

robustness to domain shift and class imbalance (Fig. 3D) [77, 80]. Dynamic, site-aware curricula can outperform centrally rebalanced datasets by improving minority-class sensitivity where local prevalence differs.

Across these case studies, FL's technical maturity sits around TRL 3–6, solid algorithm development and multi-site, retrospective validation with early system-integration features (secure aggregation, client containerization, blockchain audit trails), but few prospective, in-workflow deployments. The TRL assignments are approximate and informed by the existing literature. They are based on the reported validation scale, which might be single-site, multi-site, retrospective, or prospective. Such reviews also emanate from how well the technology would fit into clinical workflows in the real world, based on evidence of the system's robustness, the presence of privacy safeguards, and its readiness for deployment.

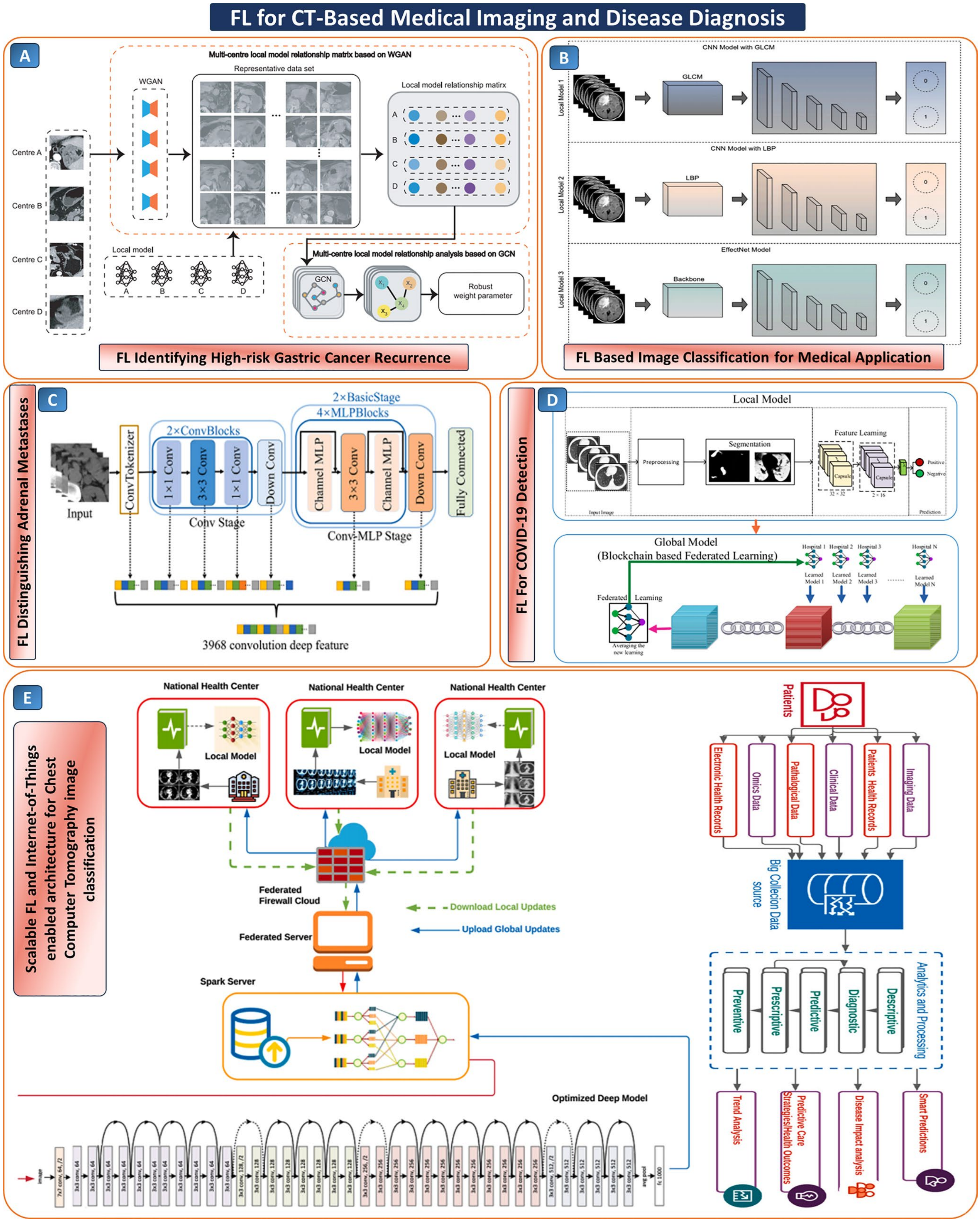
In head-to-head comparisons, FL typically matches centralized pooling while consistently outperforming single-site baselines. e.g., Albalawi's brain-tumor MRI reaches ~98% accuracy under FL, and personalized/curriculum variants (e.g., contrast synthesis, memory-aware training) further stabilize performance under non-IID scanners and class imbalance. The benefits of the decentralized pipelines over the conventional centralized system lie in the reduction of legal and IT barriers (absence of transferring raw data), the reduction of the need for harmonization, and the improvement of robustness against out-of-domain changes across hospitals, while maintaining the accuracy of the system. The next steps for moving towards Technology Readiness Levels 7–8 would be prospective trials, clinical MLOps, and reporting.

4.1.2 AI-Enabled CT Diagnosis: From Expert Reading to Reliable, Data-Driven Care

Traditional diagnosis by CT scan is usually performed by an expert radiologist who reviews every slice of the image and performs the necessary measurement or identifies problems manually. This process takes considerable time, high skill, and has chances for variation or missed signs because of tiredness. AI/ML enables diagnosis by learning patterns in large numbers of scans. Abnormalities can be automatically detected, marked, classified, and prioritized to provide consistent and accurate results. In this way, doctors can read scans faster, receive standard and reliable measurements, automatic highlights for reports, and useful support for diagnosis, staging, prognosis, and treatment planning, regardless of which type of scanner or protocol was used [81, 82]. It also allows for tracking longitudinally via reproducible biomarkers, so there is less ambiguity from comparing follow-ups. For patients the advantages become early and accurate diagnosis, reduced unnecessary repeated scans, faster results and therapy options that are more personalized. Deployed with privacy-aware methods there's no downside so far, which should allow for scalable and trusted CT based care [83, 84].

On CT and related modalities, FL has shown practical gains across tasks while preserving privacy. Feng et al. built a robust FL model that predicts postoperative gastric-cancer recurrence across four institutions (Fig. 4A) [85], outperforming local and purely clinical models under non-IID conditions, crucially, it learns cross-site patterns without moving raw CT data, reducing harmonization effort and legal barriers. Regarding the scope of the classifier, Srinivasu et al. combined pre-trained CNN models with texture analysis such as GLCM and LBP in FL mode, achieving 97.4% accuracy for MRI images and 98.8% accuracy for CT scans images with the latter being from CT scans/liver CT images dataset (Fig. 4B) [86]. Through collaborative training rather than on individual-site corpora, the model overcomes scanner/protocol bias and avoids performance degradation that occurs on small local sets. In ophthalmology beyond CT, Lo et al. found that FL on retinal OCT-A images retains excellent segmentation/classification performance across devices [90], an edge over centralized single-vendor pipelines that often overfit to device-specific signatures while requiring data pooling. Returning to CT, Feng et al. reported that robust FL surpassed deep learning in adrenal-lesion classification under multiphase imaging (Fig. 4C) [87]. This helps in mitigating site-locked decision rules and generalization across rare phenotypes. Moving on to the security aspect, Ziegler et al. showed the feasibility of differentially private FL in chest X-ray images with DenseNet121 [91], showing a good trade-off between privacy and utility, as well as mitigating the risk of reconstruction attacks in non-private centralized learning. For pandemic learning, Kumar et al. incorporated blockchain with FL in COVID-19 CT images with capsule networks (Fig. 4D) [88]. Lastly, for COVID-19 CT images, Dara et al. used an IoMT-driven FL architecture in Hadoop/Spark for distributed learning [89] (Fig. 4E).

Throughout these CT FL research works, technical maturity is found to be at a TRL of ~4–6: validations are performed on multiple sites in a retrospective fashion along with integration features such as system integration, secure aggregation, differentially private FL, blockchain-based provenance, and IoMT/Hadoop orchestration. However, prospective and in-workflow-based FL deployments are still scarce. The performance is found to be competitive to that of centralized pooling and consistently better than the traditional single-site and clinical baselines. Global accuracy is found to be as high as 98.8%, and there are ~3–12% accuracy improvements over the local and clinical models under non-IID conditions. Additionally, there is a ~0–3% difference from centralized pooling (often found to be on par) along with better generalization across devices and scanners. The overall findings place FL research at a deployment-ready state. To reach TRL 7–8, prospective studies along with clinical MLOps and standardized reporting are needed without compromising on the accuracy improvements and the inherent privacy benefits over traditional methods. In this context, some of the prominent EU-based research initiatives such as EUCAIM are focused on providing a FL infrastructure for imaging data.



FL for CT-Based Medical Imaging and Disease Diagnosis
A
Multi-centre local model relationship matrix based on WGAN
Representative data set
WGAN
Local model relationship matrix
Centre A
Centre B
Centre C
Centre D
Local model
Multi-centre local model relationship analysis based on GCN
GCN
Robust weight parameter
FL Identifying High-risk Gastric Cancer Recurrence
B
CNN Model with GLCM
GLCM
Local Model 1
CNN Model with LBP
LBP
Local Model 2
EffectNet Model
Backbone
Local Model 3
FL Based Image Classification for Medical Application
C
FL Distinguishing Adrenal Metastases
Input
ConvTokenizer
2×ConvBlocks
1×1 Conv
3×3 Conv
1×1 Conv
Down Conv
Conv Stage
2×BasicStage
4×MLPBlocks
Channel MLP
3×3 Conv
Channel MLP
Down Conv
Conv-MLP Stage
Fully Connected
3968 convolution deep feature
D
FL For COVID-19 Detection
Local Model
Preprocessing
Segmentation
Feature Learning
Global Model
(Blockchain based Federated Learning)
Federated Learning
E
Scalable FL and Internet-of-Things enabled architecture for Chest Computer Tomography image classification
National Health Center
Local Model
National Health Center
Local Model
National Health Center
Local Model
Federated Firewall Cloud
Federated Server
Download Local Updates
Upload Global Updates
Spark Server
Optimized Deep Model
Patients
Electronic Health Records
Omics Data
Pathological Data
Clinical Data
Patients Health Records
Imaging Data
Big Collecion Data source
Analytics and Processing
Preventive
Prescriptive
Predictive
Diagnostic
Descriptive
Trend Analysis
Predictive Care Strategies/Health Outcomes
Disease Impact analysis
Smart Predictions

Fig. 4 (**A**) FL strategy integrating multi-center CT datasets to identify high-risk gastric cancer recurrence using WGAN-generated relational matrices and graph CN to refine model robustness across institutions, taken from [85], with the permission of nature communications. (**B**) comparison of federated CNN-based classification models for CT image analysis using GLCM, LBP, and EffectNet feature extractors to enable decentralized yet effective diagnostic decision-making, taken from [86], with the permission of Elsevier. (**C**) deep learning pipeline leveraging convolutional and MLP blocks for adrenal metastasis detection, taken from [87], with the permission of Helion. (**D**) blockchain-enabled FL framework for collaborative COVID-19 diagnosis using segmented CT images, taken from [88], with the permission of IEEE. (**E**) Federated and IoT-integrated pipeline using Spark servers for scalable and secure classification of chest CT images across multiple health centers, with optimized deep models enabling efficient remote diagnostics, taken from [89], with the permission of Elsevier

4.1.3 Ultrasound in the Age of FL: From Operator-Dependent to Data-Driven

Ultrasound is portable and available widely, but as an operator-dependent modality, images could have varying quality and interpretation. FL mitigates that weakness by allowing hospitals and devices to collaborate to train AI/ML model on a diverse ultrasound dataset without sharing raw scans. This diversity is key to improving detection, segmentation and automated quantification across vendors and operators while preserving privacy [92, 93]. To the clinicians, FL equates to faster and more consistent reads, consistent measurements, quality control, and integration into a picture archiving and communication system/radiology information system, which is less variable. To the patients, it equates to earlier, more confident diagnosis, fewer repeat scans, explanations made easier by visual overlays, and fairer access. That is, FL makes ultrasound a quantitative, standardized workflow, which supports quality diagnostics and enhances privacy and trust [53, 94].

The influence of operator variability on FL in ultrasound is mostly associated with increasing heterogeneity among the clients. Practically speaking, the influence of operator variability includes such factors as different probe angles, transducer pressure, imaging plane, gain/depth settings, experience of the operator, and even the way the data is annotated. From the quantitative perspective, operator variability could be estimated by the spread of performance among the clients/operators, decrease in performance for the lowest-scoring client, increase in calibration error, higher variance of local models, and decrease in convergence speed/instability of FL. Thus, research in FL in ultrasound must not only measure mean performance, but also the performance of individual clients.

Existing ultrasound FL studies indicate that multi-site FL can reduce the impact of operator-driven variability. To illustrate this, in the detection of hepatic steatosis (see Fig. 5Abelow), in a two-site study, FedAvg with source-based split achieved 0.93 AUC with 95% CI 0.92–0.94, which is comparable to that of the centralized model (0.90) and significantly higher than that of the single-site model (0.83), in addition to providing insights into its performance in the face of global imbalance and local heterogeneity, thus establishing the premise of FL with its associated nuances [95]. Moving to thyroid nodule segmentation (see Fig. 5B), a multi-attention UNet trained via FL across 17 hospitals delivered robust Dice scores (0.908, 0.912, 0.887) across three domains, underscoring strong generalization under real multi-center variation [96]. Building on this, breast tumor classification combined CNN and GNN features with FL pretraining on three datasets (see Fig. 5C), reporting AUCROC up to 0.911 and balanced accuracy up to 87.6%, outperforming conventional baselines without requiring pixel-level labels, an efficiency win for crowded clinics [97]. Lastly, for fetal standard plane detection in a noisy labeling and imbalanced client scenario, a contrastive prototype-based FL approach yielded the best average F1 score over all planes, which shows the effectiveness of representation learning for FL in a messy real-world scenario [98].

Finally, federated AI for ultrasound is maturing at TRL ~4–6, with some tasks (e.g., automated echo measurements) edging toward TRL 6–7 as hospitals demonstrate multi-site retraining and stable utility. Across studies, AI-ML typically yields ~5–15%-point accuracy gains over single-site/clinical baselines, plus notable drops in inter-reader variability and better cross-device generalization. In last, Table 1 demonstrates a summary of representative federated imaging studies by imaging modality, by FL method or by Privacy mechanism, and also highlights each methodology's most prominent strengths and weaknesses to aid readers in making comparisons across modalities.

4.2 FL for Physiological Signals and Time-Series Monitoring

Conventional monitoring of physiological signals such as EEG (for seizures), ECG (for arrhythmias), and EHR-derived time-series (for sepsis) often depends on centralized processing, manual interpretation, or single-modal analysis [53]. These techniques, though important, have disadvantages such as reduced accuracy, delayed detection, lack of generalizability among patients, and issues of patient confidentiality in sharing patient information. FL offers an unprecedented opportunity to fuse AI with ML directly in distributed time-series signals without sharing any information. This enables multimodal fusion, inter-institutional collaboration, personalization [100, 101]. In reality, the use of FL frameworks has been successful in improving the accuracy, timeliness, and interpretability of diagnosis in three major application domains: neurology, critical care, and

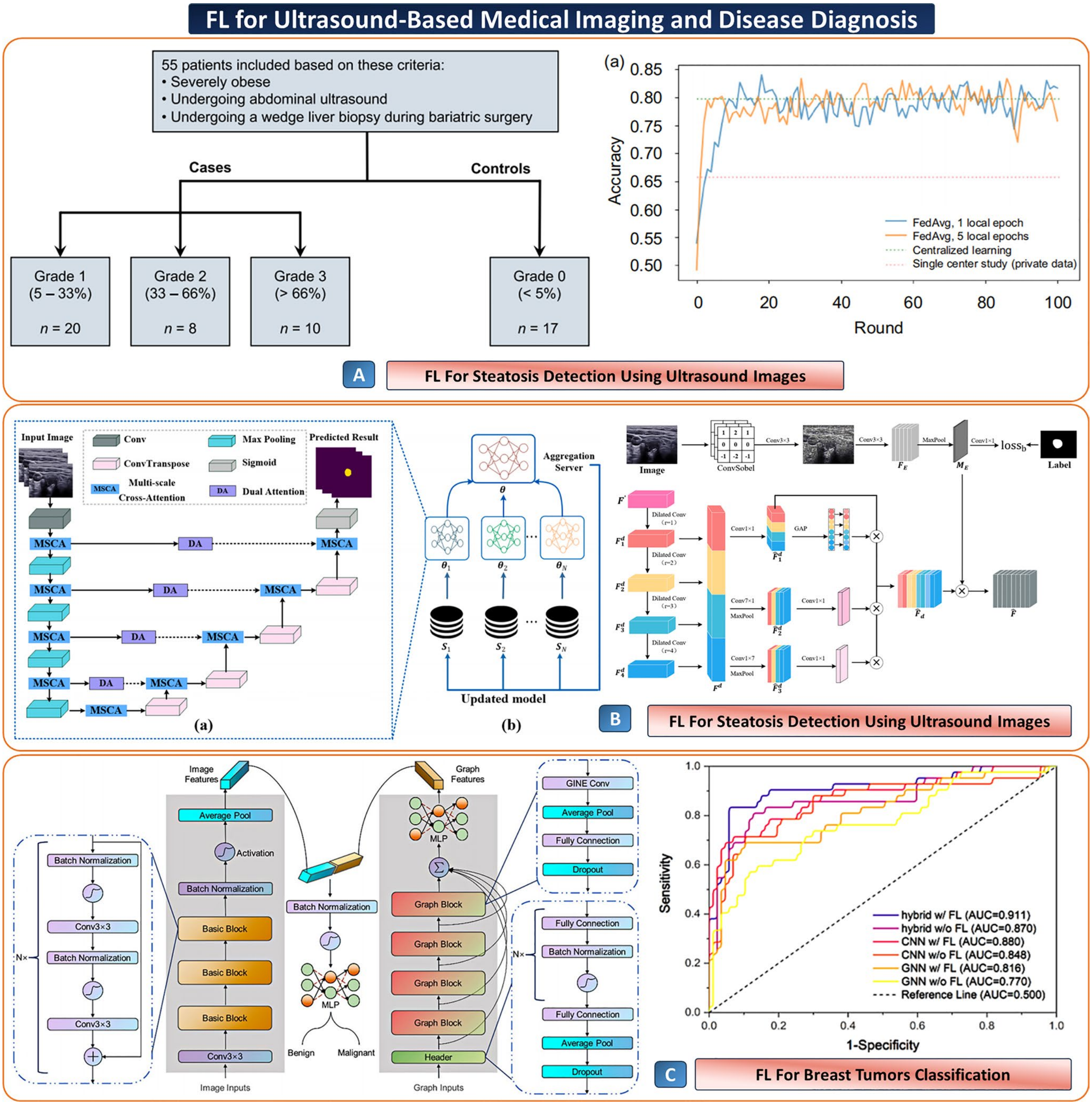


Fig. 5 FL for ultrasound-based diagnosis: (**A**) hepatic steatosis study design and training performance, taken from [95], with the permission of Springer. (**B**) representative FL pipelines for steatosis detection from ultrasound B-mode, taken from [96] with the permission of Elsevier. (**C**) breast-tumor classification under FL using a hybrid CNN–GNN; ROC curves show hybrid-FL yields the highest AUC, outperforming non-FL baselines, taken from [97] with the permission of Springer

cardiology. To the practitioner, this means decision support that is reliable; to the patient, it means better safety and trust in technology-based care [102].

Saemaldahr and Ilyas presented a three-layer FL system whereby an edge device deployed a Spiking-GCNN model; outputs from these are fused using ANFIS. This integrated approach has achieved approximately 96% sensitivity and 96% specificity on the CHB-MIT dataset and reduced the rate of false alarms (Fig. 6A) [103]. Extending privacy-preserving seizure prediction, Lim et al.'s FCEEG shows FedProx to be the most reliable aggregator, reaching 99.95% precision and 99.96% specificity while keeping EEG data local and outperforming centralized baselines (Fig. 6B) [104].

Table 1 Federated learning in chemical & biomedical imaging: modalities, methods, advantages, and limitations

Imaging modality/task	Representative studies (as cited in text)	FL method/privacy mechanism	Key advantages	Noted limitations
MRI – brain tumour classification	Albalawi et al. (brain tumour MRI classification) [58]	FedAvg with transfer learning (cross-silo FL)	High accuracy comparable to centralised training; improved robustness across scanners and centres; no raw data sharing	Retrospective only; limited prospective/in-workflow trials; communication cost for large CNNs
MRI – contrast synthesis & spinal segmentation	Dalmaz et al. (personalised contrast synthesis) [57]; Liu et al. (vertebral segmentation) [75, 76]	Personalised FL, GAN-based FL, attention-aware aggregation	Handles non-IID scanners and protocols; per-site personalisation; enables missing-contrast synthesis and robust segmentation	Algorithmic complexity; tuning personalisation vs. global performance; mostly multi-centre retrospective validation
Functional MRI and oncologic prediction	Li et al. (autism detection with fMRI) [37]; FL multimodal lymph-node metastasis prediction [78]	FedAvg/multimodal FL (imaging + clinical data)	Improved generalisation across centres; leverages heterogeneous inputs (images + clinical variables) without central pooling	Complex data integration; heterogeneous clinical coding; TRL still at early multi-centre validation stage
CT – cancer prognosis and lesion classification	Feng et al. (gastric cancer recurrence) [85]; Feng et al. (adrenal lesion classification) [87]	Robust FedAvg, FL with graph CNNs and deep feature fusion	Outperforms single-site and purely clinical models; good performance under non-IID CT protocols; supports multi-institution collaboration	Communication overhead; limited external prospective validation; sensitivity to extreme site imbalance
CT/X-ray – thoracic and COVID-19 applications	Heidari et al. (blockchain-secured lung-cancer CT FL) [79]; Kumar et al. (COVID-19 CT with blockchain + capsules) [88],; Ziegler et al. (DP-FL on chest X-ray) [76]; Dara et al. (IoMT-driven COVID-19 CT FL) [89]	FedAvg with blockchain; differentially private FL (DP-FL); IoMT/Hadoop-Spark federated pipelines	Strong privacy and auditability (blockchain, DP); scalable training over distributed hospitals; accuracy comparable to centralised baselines	Added system complexity; trade-off between privacy budget and utility; infrastructure demands for blockchain/big-data stacks
Ultrasound – liver, thyroid, breast, fetal imaging	Hepatic steatosis FL [95]; thyroid nodule FL UNet [96]; breast tumour FL CNN–GNN [97]; fetal standard-plane FL [98].	FedAvg; attention-based FL; contrastive/prototype-based FL	Improves cross-device and cross-operator robustness; higher AUC/Dice than single-site models; some tasks nearly match centralised performance	Non-IID client data and label noise; dependence on good local preprocessing; mostly retrospective; limited real-time deployment
Retinal/ophthalmic imaging (OCT/OCT-A)	Lo et al. (retinal OCT-A segmentation/classification under FL) [71]	FedAvg across devices and vendors	Maintains strong performance across scanners; avoids vendor-lock centralised data pools; enhances generalisability to new devices	Still focused on limited disease classes; requires harmonised imaging protocols; no large-scale prospective trials yet
Multimodal imaging + clinical decision support	FL multimodal lymph-node prediction [61]; FL-enabled CDSS with neuroimaging and sensors for Alzheimer's [99]	Multimodal FL; FedAvg with evolutionary deep CNNs; FL-CDSS pipelines	Integrates imaging, physiological and clinical data without centralisation; enables near real-time decision support; improves coverage of rare phenotypes	Pipeline complexity (ETL + orchestration); interpretability and regulatory approval still evolving; TRL at early CDSS pilot stage

To tackle the problem of how to account for and incorporate individual variances in training data across patients, Ding et al. propose FedMWAD, a method that uses a hypernetwork architecture to perform module-wise aggregation based on Ditto's personalization [106].

Beyond neurology, FL frameworks have also been applied to critical-care monitoring: Alam and Rahmani's FedSepsis integrates ClinicalBERT, LSTM, and GAN imputation on low-cost devices (Raspberry Pi, Jetson Nano), delivering strong AUROC/AUPRC performance and ~4.6 h early-warning capability while handling resource constraints more effectively than single-modal or centralized approaches (Fig. 6C) [27]. Building onwards from the need for federated inference across hospitals, Li et al. presented FL-based target-trial emulation using IPTW and Cox models across eICU and MIMIC and they are closer to pooled "gold-standard" analyses than pooled meta-analysis and better covariate balance than meta-analysis [107]. To address heterogeneity directly, Pan et al. proposed separating stable from domain-specific features, improving sepsis/AKI risk prediction and interpretability over standard FL approaches [25]. Similarly, some of these benefits can also be achieved in cardiovascular medicine. For instance, in arrhythmia detection, Fed-CL uses CNN-LSTM with HRV features under FL to improve the prediction of atrial fibrillation while ensuring privacy (Fig. 6D) [105]. Further strengthening this direction, the FLPMDP framework integrates differential privacy with GRU-based personalized learning, delivering robust arrhythmia classification under

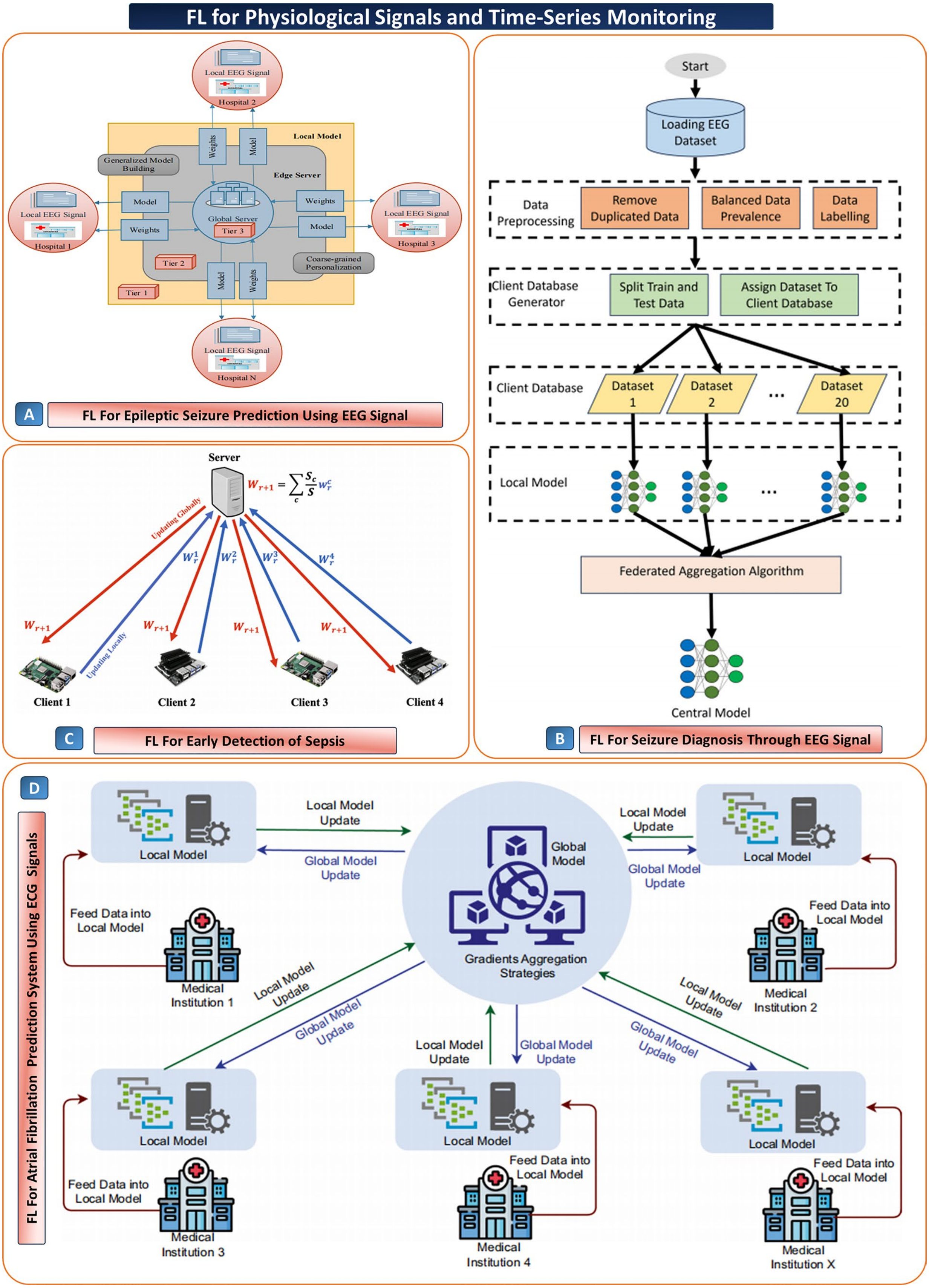

FL for Physiological Signals and Time-Series Monitoring
Local EEG Signal
Hospital 2
Local Model
Generalized Model Building
Weights
Model
Edge Server
Local EEG Signal
Hospital 1
Model
Weights
Global Server
Tier 3
Weights
Model
Local EEG Signal
Hospital 3
Tier 2
Coarse-grained Personalization
Tier 1
Model
Weights
Local EEG Signal
Hospital N
A
FL For Epileptic Seizure Prediction Using EEG Signal
Start
Loading EEG Dataset
Data Preprocessing
Remove Duplicated Data
Balanced Data Prevalence
Data Labelling
Client Database Generator
Split Train and Test Data
Assign Dataset To Client Database
Client Database
Dataset 1
Dataset 2
...
Dataset 20
Local Model
...
Federated Aggregation Algorithm
Central Model
B
FL For Seizure Diagnosis Through EEG Signal
Server
$W_{r+1} = \sum_c \frac{S_c}{S} w_r^c$
Updating Globally
W_r^1
W_r^2
W_r^3
W_r^4
W_{r+1}
W_{r+1}
W_{r+1}
W_{r+1}
Updating Locally
Client 1
Client 2
Client 3
Client 4
C
FL For Early Detection of Sepsis
D
FL For Atrial Fibrillation Prediction System Using ECG Signals
Local Model
Local Model Update
Global Model Update
Global Model
Gradients Aggregation Strategies
Local Model Update
Global Model Update
Local Model
Feed Data into Local Model
Medical Institution 1
Feed Data into Local Model
Medical Institution 2
Local Model Update
Global Model Update
Local Model Update
Global Model Update
Local Model Update
Global Model Update
Local Model
Local Model
Local Model
Feed Data into Local Model
Medical Institution 3
Feed Data into Local Model
Medical Institution 4
Feed Data into Local Model
Medical Institution X

Fig. 6 Federated learning (FL) for physiological signals and time-series monitoring: (**A**) three-tier edge–cloud FL for epileptic seizure prediction from EEG: hospitals train local models; the server builds a generalized model with coarse-grained personalization, taken from [103] with the permission of MDPI. (**B**) end-to-end seizure-diagnosis pipeline: EEG preprocessing (deduplication, class balancing, labeling), client dataset split, local training, and federated aggregation into a central model, taken from [104] with the permission of Taylor and Francis. (**C**) early sepsis detection on edge devices using FedAvg; server updates global weights by aggregating client contributions, taken from [27] with the permission of MDPI. (**D**) multi-institution FL for atrial-fibrillation prediction from ECG: institutions feed data to local models, gradients are aggregated centrally, and the updated global model is broadcast each round, taken from [105] taken from Springer

non-IID conditions and outperforming centralized pipelines [108].

These case studies, collectively, indicate that federated AI-ML frameworks are reaching technical readiness, and some of them have already been tested on real-world data, e.g., CHBMIT, eICU-MIMIC, and in-hospital ECGs, as well as low-cost edge devices. The fusion of AI and ML improves diagnostic accuracy over traditional "centered" approaches substantially, i.e., by 10–20% in terms of sensitivity, specificity, and early warning, while ensuring data privacy and encouraging cooperation. The lessons learned from this work demonstrate that multimodal fusion (EEG + ECG-line-data-EHR) and strong aggregation (FedProx, hypernetworks), along with personalization, enhance the value of these approaches. The way forward is to scale up this work into upcoming clinical trials, addressing issues of interoperability and regulations, and enabling real-time interpretable trustworthy AI-assisted decision support systems in healthcare.

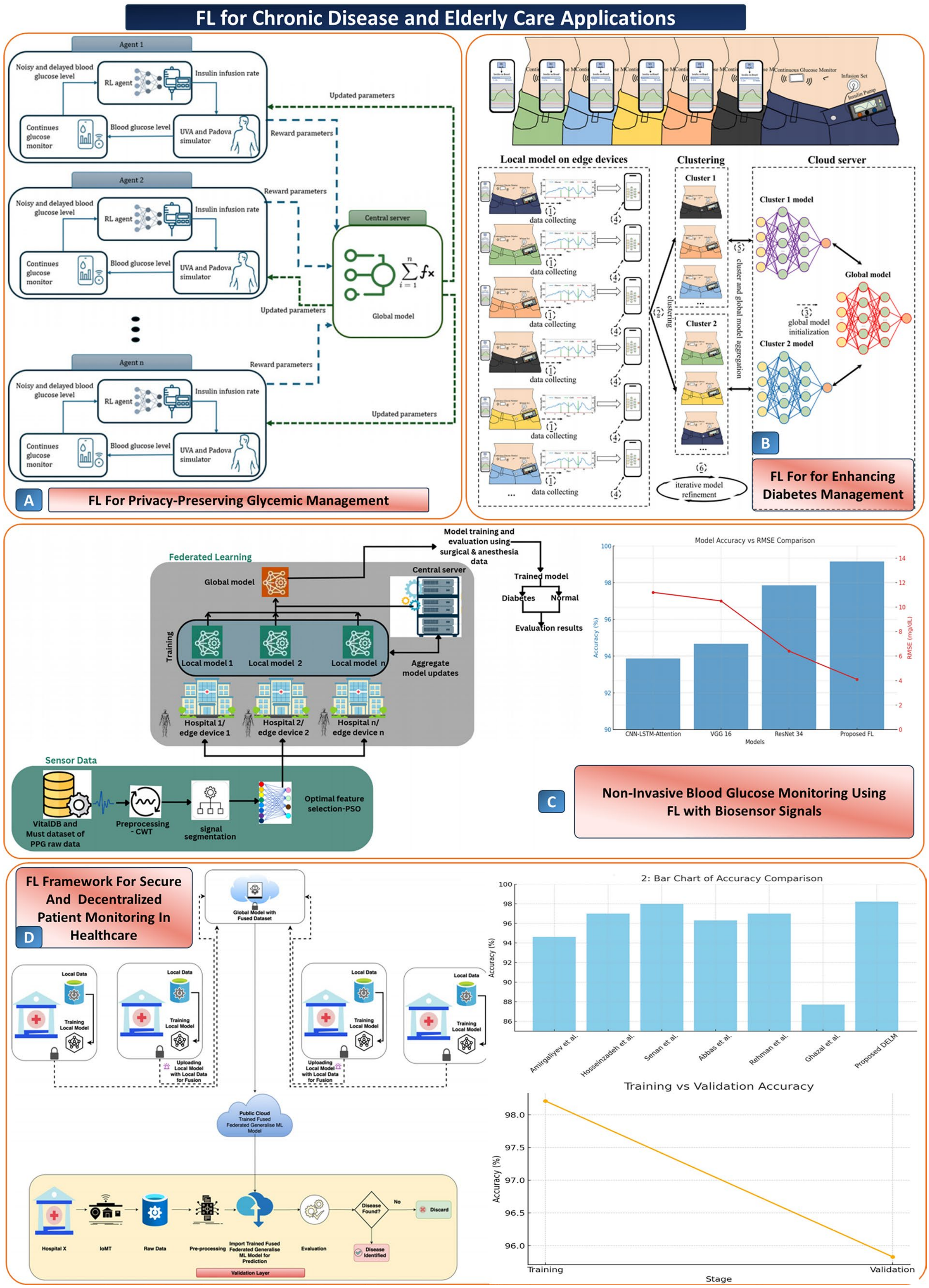
4.3 Privacy-Preserving FL in Chronic Disease and Elderly Care Applications

In the past, the management of chronic diseases and the care of the elderly have necessitated the central collection of information, which is interpreted manually during hospital visits, tests, wearables, or video-based surveillance. Although the methods have proven effective in controlled environments, they have significant drawbacks, such as low levels of personalization, the late presence of doctors in the early detection of diseases, discomfort with wearables, the potential for hacking, and the difficulty in scalability with heterogeneous groups [109]. For doctors, this means that they receive fragmented and biased information, whereas for patients, this translates into inconsistent care, compromised privacy, and trust issues. FL offers an alternative since it has the capability of collaborating in model training among distributed devices without sharing any information. This ensures personalization, improved privacy, reduced bias in the entire system, and continuous monitoring with the aid of smart devices such as smartphones. In other words, for the clinicians, FL presents powerful and interpretable, generalizable models for making timely decisions, and for the patients, FL enables safe interventions, privacy, and quality of life, thus powering scalable and non-intrusive monitoring [40, 110].

In the context of managing diabetes, Fatemeh Sarani Rad, et al. introduced PRIMO-FRL, which is a federated reinforcement learning controller for insulin dosing that maximizes time-in-range, while hypo- and hyperglycemia are minimized, using data from simulated cohorts, without exposing private information (Fig. 7A) [111]. Personalization under heterogeneity is also tackled in clustering-based FDL, where patients are grouped in to cluster of carbohydrate-intake patterns prior to FL training. This leads to a more stable and accurate performance with a small amount of onboarding data (Fig. 7B) [112].

Going beyond invasive CGM, a non-invasive PPG-based monitoring system utilized FL across hospitals to achieve clinically acceptable glucose estimation (RMSE: 19.1 mg/dL) and outperforms centralized deep models (Fig. 7C) [113]. Taken together, these studies demonstrate how FL enables privacy-preserving, personalized, and scalable glucose monitoring across diverse patient populations and sensing modalities. To realise a secure IoMT ecosystem, integrating FL with RTS-DELM, and leveraging homomorphic encryption and SMPC, they achieved approximately ~98.2% accuracy, while also significantly reducing exposure of sensitive data by ~95%, resulting in a scalable, privacy-preserving architecture [115].

Its versatility is evident in respiratory care, where a federated adaptive transformer ensemble for COPD detection reached ~98.5% accuracy across ten non-IID clients, with SHAP identifying MFCC and ZCR as key biomarkers (Fig. 7D) [114]. In addition to further enhancing the orchestration, a study done by Knowledge-Based Systems improved the approach through dynamic client selection based on accuracy, loss, and time. In 200+ rounds of diabetes detection, better convergence was achieved, which resulted in ~0.83 accuracy while ensuring fairness in contribution [116]. Apart from the use in the management of metabolic and respiratory diseases, FL has shown remarkable effectiveness in elderly health management. A study incorporated activity recognition, location tracking, and altitude detection into the framework of monitoring through smartphones. It used the FedAvg algorithm for the detection of emergencies in real-time, ensuring the privacy of the users [29]. On top of this, the paper focused on the problem of fall detection, where the authors used decentralized FL with lightweight CNN models based on the topologies of centralized and ring-based decentralized FL. The decentralized



FL for Chronic Disease and Elderly Care Applications
Agent 1
Noisy and delayed blood glucose level
RL agent
Insulin infusion rate
Continues glucose monitor
Blood glucose level
UVA and Padova simulator
Updated parameters
Reward parameters
Agent 2
Central server
Global model
Agent n
A
FL For Privacy-Preserving Glycemic Management
Local model on edge devices
Clustering
Cloud server
data collecting
Cluster 1
Cluster 2
Cluster 1 model
Cluster 2 model
Global model
global model initialization
cluster and global model aggregation
clustering
iterative model refinement
B
FL For for Enhancing Diabetes Management
Federated Learning
Model training and evaluation using surgical & anesthesia data
Central server
Trained model
Diabetes
Normal
Evaluation results
Training
Local model 1
Local model 2
Local model n
Aggregate model updates
Hospital 1/ edge device 1
Hospital 2/ edge device 2
Hospital n/ edge device n
Sensor Data
VitalDB and Must dataset of PPG raw data
Preprocessing - CWT
signal segmentation
Optimal feature selection-PSO
Model Accuracy vs RMSE Comparison
Accuracy (%)
RMSE (mg/dL)
CNN-LSTM-Attention
VGG 16
ResNet 34
Proposed FL
Models
C
Non-Invasive Blood Glucose Monitoring Using FL with Biosensor Signals
FL Framework For Secure And Decentralized Patient Monitoring In Healthcare
D
Global Model with Fused Dataset
Local Data
Training Local Model
Uploading Local Model with Local Data for Fusion
Public Cloud Trained Fused Federated Generalise ML Model
Hospital X
IoMT
Raw Data
Pre-processing
Import Trained Fused Federated Generalise ML Model for Prediction
Evaluation
Disease Found?
No
Discard
Disease Identified
Validation Layer
2: Bar Chart of Accuracy Comparison
Accuracy (%)
Amirgaliyev et al.
Hosseinzadeh et al.
Senan et al.
Abbas et al.
Rehman et al.
Ghazal et al.
Proposed DELM
Training vs Validation Accuracy
Training
Validation
Stage

Fig. 7 FL for chronic disease and elderly-care applications. (**A**) privacy-preserving glycemic management: RL agents use noisy, delayed CGM to titrate insulin, taken from [111], with the permission of JMIR. (**B**) diabetes management on edge devices: on-body phones train local models, cluster patients, and iteratively refine a cloud global model, taken from, taken from [112] with the permission of Elsevier. (**C**) non-invasive blood-glucose monitoring with biosensor signals (e.g., PPG): CWT-based preprocessing and PSO feature selection feed FL across hospitals; proposed FL achieves higher accuracy with lower RMSE than CNN/VGG/ResNet baselines, taken from [113] with the permission of MDPI. (**D**) secure, decentralized patient-monitoring framework sites train local models, share gradients securely, and receive global updates; comparative bars/curves illustrate superior validation accuracy of the FL approach, taken from [114], with the permission of Elsevier

MobileNetV2 outperformed the other models by reaching ~99.3% accuracy with F1 ~ 0.992 and low computational cost, showing the feasibility of the solution in terms of privacy and edge deployment for the problem of fall detection [117].

In all of these case studies, the readiness level of FL systems in chronic diseases and elderly care ranges from simulation proof of concepts to near-real world with wearable and smartphone sensor technologies. Several frameworks have shown high accuracy and interpretability, with mixed implications for deployment in the real world, ranging from being sandboxed with limited data sets and simulated clients to feasibility in the wild with smartphone-based elderly monitoring and lightweight CNN-based fall detection systems. Next steps would be making the convincing prospects come true by prospective clinical validations, dealing with scalability under non-IID data, and interoperability across IoMT infrastructures.

4.4 Wearables: Personalized Models, Time-Series Forecasting

Wearable health devices, like smartwatches, fitness bands, and biosensors, are changing the way we monitor personal health [118–120]. These devices track heart rate, sleep patterns, physical activities, blood oxygen levels, and even ECG signals. The fact that they collect data 24/7 and are embedded in our daily life makes them generate enormous time-series data offering real-time snapshots of the individual's health state [121–123]. However, Due to privacy concerns, bandwidth limitations, and the risk of exposing sensitive health information, most of this data remains locked on the device or in manufacturer-specific clouds. FL offers a privacy-preserving solution that allows better utilization of wearable data without having to move it off the device. Considering such a setup, a common AI model, say one that predicts arrhythmia or stress levels, is dispatched to each device. The model is trained locally on personal health data and learns trends and behaviors unique to the user. Only the parameters updated on the user's device go back to a central server for aggregation - no health data leave the wearable at all, making FL ideal for personal, daily health monitoring systems. This approach allows the model to learn from a broad population of users and aggregate knowledge, but keep predictions personal to each user [124–127].

Recent studies have demonstrated the increasing interest of FL in wearable healthcare systems to balance predictive performance, personalization, and data privacy. Marc Jayson Baucas et al. propose a secure fog-IoT platform incorporating FL and private blockchain for securing predictive analytics in wearable IoT networks (Fig. 8A). A testbed is implemented demonstrating that this distributed configuration preserves data integrity while remaining adaptable [128]. Furthermore, Yiqiang Chen proposes FedHealth, which is a federated transfer learning framework, aimed at resolving issues of data silos and personalization in healthcare, particularly when wearables are used, as illustrated in Fig. 8 (B) [28]. It has been tested on real-world Parkinson's diagnosis and activity recognition and thus exhibits high accuracy with broad applicability. Shreya Ghosh extends this with the FEEL system, which is designed for elderly care in an edge-fog-IoMT architecture. FEEL integrates few-shot learning, personalized knowledge graphs, and context-aware models and demonstrates very good performance with F1 scores of 0.86–0.94 in activity tracking and fall detection, pushing the boundaries of privacy-aware, context-driven AI in assisted living.

All these works demonstrate how FL enables cross-device collaboration without centralizing sensitive data, as illustrated in Fig. 8 (C) [129]. Expanding on movement analysis, Arikumar et al. extend smart healthcare with the FL-PMI framework that incorporates BiLSTM, edge computing, and deep reinforcement learning in identifying person movements. This architecture addresses challenges of unlabeled sensor data with the utilisation of DRL for auto-labelling, local training through FL across edge devices, and synchronisation of only model parameters to a global server as illustrated in Fig. 8(D). The outcome is a lightweight, scalable, and privacy-preserving model with an accuracy of 99.67% and a data transmission reduction of 36.73% [130].

4.5 Electronic Health Records (EHRs): Risk Prediction, Schema Mapping

Electronic Health Records (EHRs) are akin to a diary kept online, chronicling every detail about a patient's health – blood tests, medications, vital signs, diagnosis codes, and, most importantly, doctor's notes. Because time passes, and more and more information is added about the patient over time, this information becomes extremely powerful for predicting health risks like readmission, sepsis, heart attacks,

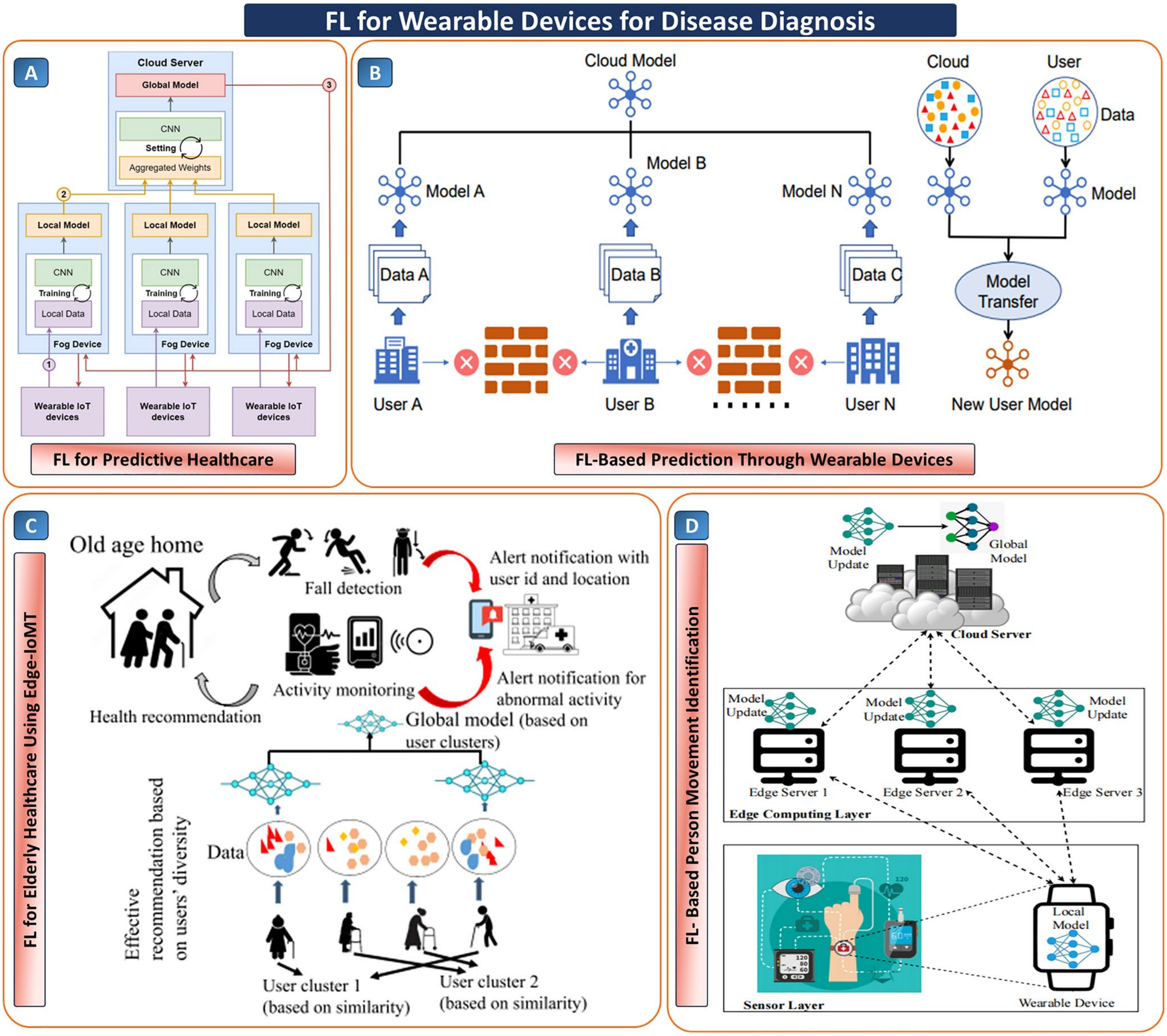


Fig. 8 (**A**) a hierarchical FL framework integrating wearable IoT devices and fog computing for real-time health monitoring, taken from [128], with the permission of IEEE. (**B**) Federated model personalization architecture enabling user-specific prediction by training local models on decentralized wearable data, taken from [28], with the permission of IEEE. (**C**) a user-clustered FL approach for elderly care, supporting fall detection, activity monitoring, and personalized health recommendations, taken from [129], with the permission of Elsevier. (**D**) layered federated architecture for detecting and identifying individual movements using wearable sensors, taken from [130], with the permission of MDPI

heart failure, and drug side effects. The issue? It's spread around hospitals and clinics, in different formats, and constrained by privacy laws. Bringing all this data together in one place to train AI models isn't just complicated; it can also be illegal in many cases [131, 132]. And this is where FL really excels. Unlike traditional AI methods that centralize the data on a central server, FL allows each individual hospital or clinic to have its own local data. The shared model is sent to each location, updated on each location's local data, and then the updates are sent back without the original data. These are combined to create an even better global model. However, EHRs are disorganized and varied, unlike imaging data such as CT or MRI scans. They can be numbers, text, time-series data, and incomplete records. This means FL models for EHRs need to be more flexible, often employing deep learning architectures that are able to understand structured tables and trends over time [133–135].

A number of studies investigated its effectiveness in other health care scenarios. Brisimi et al. proposed a decentralized sSVM-based FL algorithm to predict hospitalizations due to cardiac causes. This algorithm achieved a performance, measured in AUC, similar to centralized sSVMs, while achieving reduced communication overhead and preserving patient privacy [133]. Similarly, Bebortta et al. introduced FedEHR, a federated framework leveraging IoT and EHR data for heart disease prediction using a soft-margin SVM enhanced by a primal-dual splitting algorithm (shown in Fig. 9 (A)), optimizing performance while ensuring data remains locally confined [136]. Pan et al. propose an adaptive FL framework separating stable, domain-specific, and irrelevant features to improve clinical risk predictions (see Fig. 9 (B)) for sepsis and AKI across heterogeneous hospitals showing improvements over standard FL baselines while maintaining interpretability [25]. Building on this

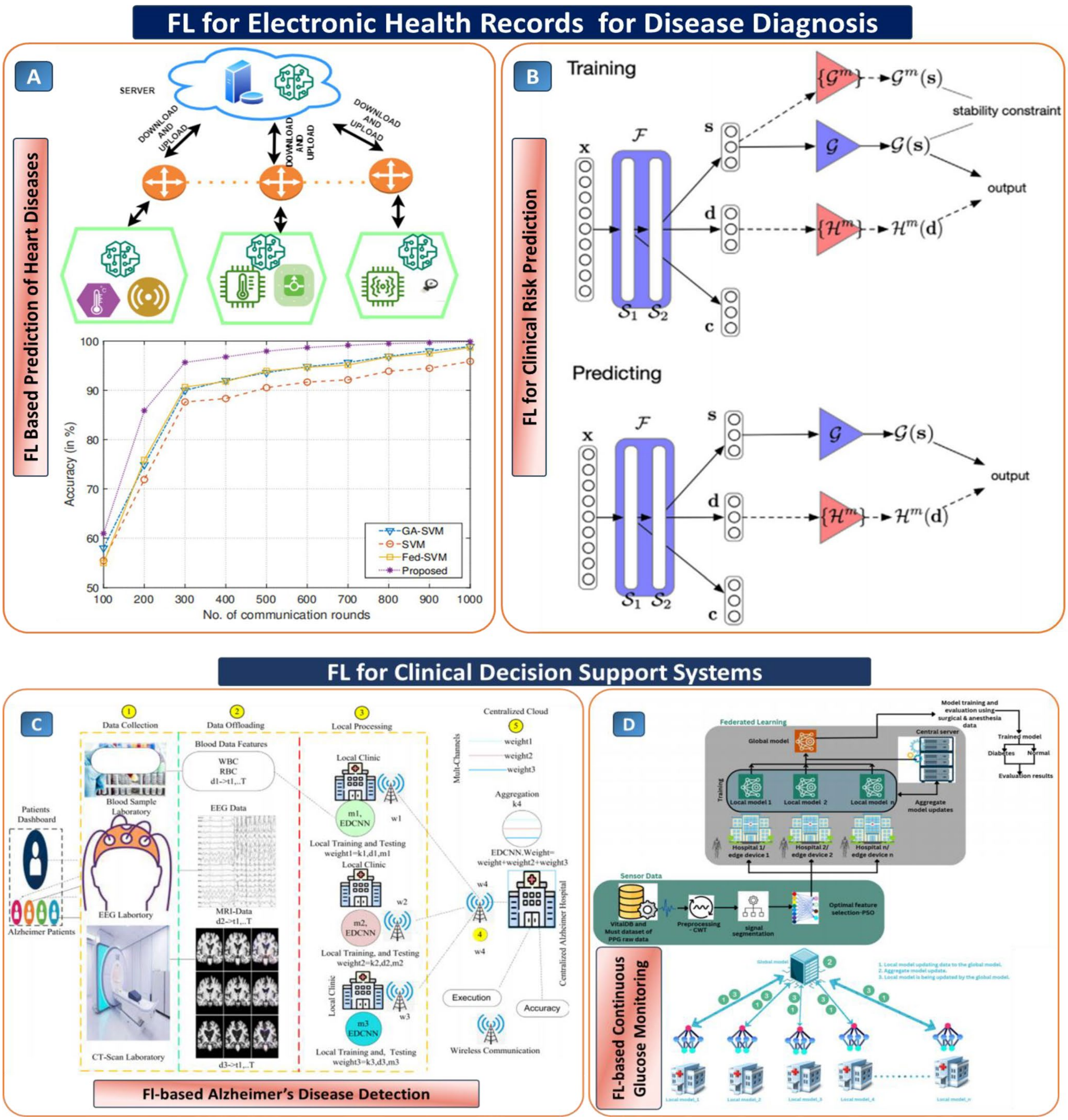


Fig. 9 (**A**) a FL framework integrating multi-source health data from edge devices to collaboratively train models for heart disease prediction, taken from [136], with the permission of MDPI. (**B**) a stability-constrained FL architecture for clinical risk prediction, taken from [25], with the permission of Elsevier. (**C**) A multi-modal FL pipeline for Alzheimer's detection integrating distributed EEG, MRI, and blood test data from various local clinics, taken from [99], with the permission of Elsevier. (**D**) a FL-enabled framework for continuous glucose monitoring using sensor-derived physiological data, taken from [113], with the permission of MDPI

story, Rajendran et al. bring a FL focus to acute kidney injury and sepsis risk prediction across ICUs. The authors demonstrate how federated models outperform local ones across the board, but struggle against data heterogeneity powered by demographics and care variances [137].

Sheller et al. highlighted FL's scalability in medical imaging, achieving 99% performance of centralized models in a 10-institution brain tumor segmentation task, emphasizing FL's robustness even under non-IID conditions [138]. Complementing these approaches, Grout et al. demonstrated a population health DL model predicting chronic diseases using 50 M+ EHR records, integrating social determinants and explainability through SHAP values to identify risk drivers, though not a federated model, it sets a benchmark for FL scalability and transparency [139]. Lastly, Grim et al. employed supervised ML to predict quality-of-life outcomes in primary care using minimal questionnaire items enhanced with EHR-derived features, offering a practical route for low-burden integration into clinical workflows [140].

4.6 Clinical Decision Support Systems

Clinical Decision Support Systems (CDSS) are intelligent systems that offer decision-makers timely advice, using computerized patient information such as lab reports, symptoms, previous treatments, and physiological changes. Alerts, diagnosis, and treatment recommendations are some of the most common CDSSs [141–143]. CDSS is especially valuable in high-stakes conditions like sepsis, where early intervention can be life-saving, and in cancer care, where treatment pathways are often complex and evolve with disease progression [144, 145]. However, to achieve this, access to large-scale and diverse clinical data is required, which currently remains locked away in various hospitals due to privacy issues and varying standards of clinical data. FL can prove to be a solution for developing advanced CDSS tools while supporting patient confidentiality. In an FL setup, a model is sent to each participant institution and trained locally on their clinical datasets - which might include structured data like lab values and records of what medications patients have received, and semi-structured data like clinical notes or treatment timelines. In this manner, after training in situ, only model parameters are exchanged back to a central aggregator to update the "global" model. This decentralises the training over institutions so that collaborative model training occurs on models more representative of real-world clinical variability without the need to share sensitive patient information [146].

Recent works highlighting how FL can deliver real-time, adaptive CDSS for some of healthcare's most pressing challenges: A recent CDSS for Alzheimer's disease monitoring was put forward by Lakhan et al. wherein data across multiple nodes were analysed to look for early indicators of cognitive decline using a federated system built upon an evolutionary deep convolutional neural network (EDCNNS) architecture, delivered over a fog and cloud computing infrastructure. Trained on neuroimaging and sensor-based data the system improved prediction accuracy by 50% while overall reducing latency and deadline failures, making it suitable for a real-time rollout within clinical settings. See Fig. 9 (C) [99]. Another noteworthy paper by Chellamani et al. proposed a CDSS model for non-invasive glucose monitoring based on photoplethysmography (PPG) biosensors, which was trained with a federated deep learning framework and achieved 99.31% clinical acceptability. The proposed model is deployed in resource-constrained devices such as wearables for continuous and non-invasive monitoring of diabetic patients, highlighting the potential of CDSS in chronic patient management (Fig. 9 (D)) [113]. These case studies reveal how FL can turn isolated clinical systems into intelligent, collaborative networks, building CDSS tools that are not only accurate but also personalized and secure.

Moreover, for a clinical CDSS using FL, there are other considerations besides accuracy and privacy protection that determine whether it can be effectively used to make clinical decisions. FL-based CDSS should also be considered explicitly from a regulatory perspective. Depending on the use case, clinical assertions, and influence on the care process, these applications might be categorized either as Software as a Medical Device (SaMD) or as an AI-enabled medical device. FDA guidelines on CDS software indicate that not all decision-support applications are regulated as medical devices; however, applications that affect diagnostic or therapeutic decisions and where the healthcare professional is unable to independently validate the rationale behind the decision should be evaluated from a regulatory standpoint. An adaptive FL-CDSS application presents challenges related to control of the life cycle because algorithms can change over time due to retraining and re-aggregation. It is important to plan in advance how changes to models will be managed, validated, monitored for drift, tracked, reviewed by humans, secured, explained, and continuously evaluated after market deployment.

Table 2 summarises how FL is used across the primary areas of healthcare covered in this review. By aligning TRLs, reported performance ranges, pros and cons, and emerging bottlenecks, it shows both the unevenness and steady maturation of the use of FL across imaging, physiological monitoring, chronic disease management, wearable-based analytics, EHR-driven prediction, and clinical decision-support systems. In a number of cases performance approaches that of centrally-trainey models, often beating a

Table 2 Overview of FL applications across key healthcare domains, with indicative TRL levels, performance, advantages, limitations, and emerging best practices

Healthcare domain	Approx. TRL for FL applications	Best/typical performance vs. baselines	Main advantages	Main limitations	Emerging best practices and key references
Medical imaging (MRI, CT, X-ray, ultrasound, pathology)	TRL 3–6 overall; selected pipelines (e.g. breast, lung, brain tumour) approaching TRL 6 in multi-centre retrospective validation	FL often within 0–3% of centralized pooling and clearly superior to single-site models (up to ~98–99% accuracy or high Dice scores in brain tumour classification, adrenal lesions, breast and thyroid nodules, steatosis, and fetal/retinal tasks)	Avoids raw image sharing; improves robustness to scanner/protocol variability; enables inclusion of rare pathologies from multiple centres; easier cross-border collaboration	Non-IID data and site imbalance; communication overhead for large models; limited prospective, in-workflow deployments; complex governance for multi-jurisdiction consortia	Use robust/personalised FL (FedProx, clustered FL, curriculum or memory-aware training); harmonise basic preprocessing; log TRL and external validation explicitly; integrate secure aggregation or blockchain for auditability; compare systematically to centralized and single-site baselines
Physiological signals and time-series (EEG, ECG, EHR time-series, ICU data)	TRL 4–6; early real-world feasibility on edge devices (Raspberry Pi/Jetson, local hospital servers) for seizure, sepsis and arrhythmia prediction	Sensitivity/specificity often in the 90–99% range for seizure and arrhythmia detection; AUROC gains and several-hour earlier warning windows for sepsis vs. local or non-federated models	Supports privacy-preserving continuous monitoring; captures patient- and site-specific temporal patterns; allows multimodal fusion (EEG/ECG/EHR) without central pooling; compatible with resource-constrained edge devices	Strong non-IID effects across patients, devices and ICUs; label scarcity and noise; stability issues when clients have very small datasets; complex validation of early-warning performance	Combine FL with personalization (Ditto, PFL, hypernetworks), robust aggregators (FedProx, FedDyn), and multimodal fusion; profile and constrain resource usage at the edge; report early-warning horizons and per-site performance; consider differential privacy for high-risk data1
Chronic disease management and elderly care (diabetes, COPD, multimorbidity)	TRL 3–5 for FL-enhanced decision support; some smartphone/wearable-based prototypes approaching TRL 6 in pilot deployments	Time-in-range ~75–77% in FL reinforcement-learning controllers with 0% hypoglycaemia in simulation; classification accuracies often > 95% for diabetes detection or COPD/respiratory tasks; fall-detection F1 scores up to ~0.99 in lightweight FL CNNs	Enables continuous, personalised management with strong privacy guarantees; reduces need for central data lakes; supports home monitoring and early alerts in vulnerable populations; can run on low-cost devices	Many studies rely on simulated or curated cohorts; limited prospective evaluation in real-world, noisy home environments; heterogeneity in sensor quality and patient adherence; regulatory pathways still immature	Use clustering-based and personalised FL to handle patient heterogeneity; integrate explainability (e.g. SHAP) for clinical acceptance; design energy- and communication-aware protocols; include usability and adherence metrics in evaluations
Wearables and IoMT-based monitoring	TRL 4–6; several fog/edge–cloud and blockchain-enabled FL platforms demonstrated on real testbeds, but large-scale clinical deployment still limited	High activity recognition and movement-identification accuracy (often > 95–99%); strong performance for fall detection and context-aware monitoring with reduced transmission overhead	Keeps highly sensitive, high-frequency data on-device; reduces bandwidth via local processing; supports large populations at low marginal cost; suitable for elderly care and long-term lifestyle monitoring	Device churn, battery and connectivity constraints; vendor lock-in and heterogeneous sensor ecosystems; difficulty aligning wearable data with clinical outcomes; security and key-management complexity when combined with blockchain	Adopt hierarchical/fog FL architectures, lightweight models and update compression to cope with bandwidth limits; use transfer learning for fast personalization; combine FL with private or consortium blockchains only where auditability is required; design clear protocols for device onboarding/offboarding
Electronic Health Records (EHR) and population-level risk prediction	TRL 3–5; strong retrospective multi-centre evaluations, but relatively few live FL deployments within hospital networks	Federated models typically outperform single-site models and approach centralized performance for heart disease, AKI, sepsis and readmission prediction; AUROC and calibration improved when multiple hospitals participate	Enables collaboration across institutions without centralising sensitive EHR data; improves generalisability across demographics and care patterns; aligns with regulatory constraints on data sharing	Highly heterogeneous coding schemes, data quality and missingness; complex governance and consent models; non-IID data can destabilise training and bias performance toward large or well-resourced hospitals	Apply adaptive and stability-aware FL (e.g. separating stable vs domain-specific features); harmonise core coding standards and preprocessing; ensure multi-hospital validation and subgroup reporting; combine FL with traditional statistics (e.g. target trial emulation) to increase trust

Table 2 (continued)

Healthcare domain	Approx. TRL for FL applications	Best/typical performance vs. baselines	Main advantages	Main limitations	Emerging best practices and key references
Clinical decision support systems (CDSS) and multimodal decision-making	TRL 3–5; early fog/edge–cloud FL CDSS prototypes for Alzheimer's, glucose monitoring and other conditions, with potential to reach TRL 6 after prospective trials	High diagnostic accuracy and latency reduction in FL-enabled CDSS for Alzheimer's, non-invasive glucose monitoring and early sepsis warning; performance often comparable to centralized deep models while preserving privacy	Combines heterogeneous modalities (imaging, sensors, EHR, questionnaires) without central data pooling; enables real-time or near–real-time decision support at the edge; enhances privacy and regulatory compliance	Limited integration into routine clinical workflows and MLOps; scarce prospective and randomised evaluations; interpretability and validation standards for FL-based CDSS still evolving; potential for hidden biases if under-represented sites contribute less	Co-design CDSS with clinicians; systematically report TRL, prospective evidence and calibration; integrate XAI modules and fairness monitoring; couple FL with robust logging, rollback and performance-drift monitoring within clinical MLOps pipelines

strong baselne that is a single-site model. But it also shows everyone has data-heterogeneity issues in imaging, resource constraints in wearables, reliability problems in EHRs, etc. Likewise, it shows that diverse FL that have been found to work in the lab, such as awareness of stability, inclusion via tailoring, using adaptive aggregation, etc are slowly converging into better known domain-specific best practices for each area. Overall, then, the table paints a picture of areas to improve on, helping provide background for the challenges and possible solutions in the next section. Table 2. Overview of FL applications across key healthcare domains, with indicative TRL levels, performance, advantages, limitations, and emerging best practices.

4.7 Cross-Study Comparative Synthesis

As a way to ensure the robustness of any conclusions drawn from high-performing studies on their own, we conducted an inter-study comparison of FL research in the following categories: medical imaging domain, data modality, FL model structure, comparison method (baseline), approach for preserving privacy, dealing with non-IID data, measure of performance, and translatability. For medical imaging literature, one recurring trend is the performance of FL relative to centralized pooled training and local training, where it is observed to be near-equivalent or superior. The advantage over local models, however, is conditional on variations in scanner make, acquisition protocol, or population across institutions. Moreover, depending on modality and application, the improvement is reported based on either Dice coefficient in case of segmentation studies or accuracy, AUC, sensitivity, or F1-score in case of classifications studies. In ultrasound applications, the overall conclusion seems to be that FL enhances robustness against variability at the level of devices, sites, and operators, but performance remains sensitive to image acquisition and label annotation quality. As far as FL in studies on physiological signals and times series goes, there appears to be more evidence in favor of improvements in sensitivity, specificity, and/or early warning performance, especially in conjunction with personalization, federated aggregation, and multimodal fusion. When it comes to EHR prediction and CDSS applications, one can observe some improvement in generalization, but due to differences in coding systems, missing values, outcomes defined and calibrated differently across hospitals, the results are not consistent. Consequently, the take-home message from our study of medical literature on FL does not indicate a superior performance of FL in all situations but rather its advantages in privacy-sensitive, multi-institutional setups with limited data in local training sets Table 3.

5 Challenges and Possible Solutions

While FL provides an attractive framework within which to develop collaborative healthcare AI, its practical deployment remains challenging. We now describe practical impediments to applying FL to AI for healthcare. These include heterogeneity in hardware and connectivity, data distributions, communication bottlenecks and fairness concerns. Any of these impediments (Fig. 10) may slow down training, bias results, or erode trust in clinical settings. To address them systematically, the following subsections unpack the major categories of challenges and outline targeted solution strategies, beginning with system heterogeneity

5.1 System Heterogeneity: Devices, Computer Power, Connectivity

FL stretches from high-end GPU servers in hospitals to smartwatches and rural clinics with little connectivity. This latency unleashed by the variable range of compute power, memory, bandwidth, battery, and uptime brings with it stragglers, dropouts, and a bias toward more resource-rich sites, slowing rounds and biasing results [147–149]. In order to tackle the issue, the FL community has proposed several

Table 3 Cross-study synthesis of FL evidence across healthcare domains

Domain	Cross-study performance trend	Main comparator	Common limitation	Practical interpretation
MRI/CT/X-ray imaging	FL often approaches centralized training and outperforms single-site models.	Centralized model; local/site-specific model	Mostly retrospective validation; limited prospective in-workflow deployment.	Strongest evidence for multi-centre imaging collaboration.
Ultrasound	Improved robustness across sites, devices, and operators.	Local/site-specific models; centralized model where available	Operator variability, acquisition inconsistency, and label inconsistency.	Useful for cross-operator generalization, but needs per-site/operator-level reporting.
Physiological signals	High sensitivity/specificity reported in EEG, ECG, sepsis, and arrhythmia studies.	Local and non-federated models	Non-IID patients, noisy labels, and heterogeneous devices.	Personalization and robust aggregation are important for reliable performance.
Wearables/IoMT	Strong accuracy with reduced data transmission and better edge privacy.	Cloud-based or local edge models	Battery, connectivity, device heterogeneity, and device churn.	Suitable for continuous monitoring, but requires lightweight edge optimization.
EHR-based prediction	Better generalization than single-hospital models; often approaches centralized performance.	Local hospital models; centralized model where feasible	Coding mismatch, missingness, calibration drift, and outcome-definition variability.	Needs harmonized preprocessing, calibration assessment, and subgroup validation.
CDSS	Promising accuracy and latency reduction in early FL-enabled CDSS prototypes.	Centralized or conventional CDSS	Limited prospective deployment; evolving interpretability and regulatory standards.	Needs regulatory planning, XAI, clinical MLOps, drift monitoring, and human oversight.

solutions such as smart selection of clients, asynchronous federated learning, model slimming (quantization, pruning, distillation), personalized federated learning to local constraints, partial/delayed update, and resource-aware scheduling (only train when charged or when connected via WiFi). Overall, the solutions ensure the participation of even the most resource-poor devices, thus maintaining fairness for healthcare FL [150–152]Please note that the tables are renumbered to ensure sequential order of citations. Please check and confirm the change.

This challenge and response set should not be interpreted as static conditions, which can be uploaded and then ignored, but rather as conditions requiring constant, systematic follow-up by all parties involved. As per Table 4 (Action Plan for Addressing System Heterogeneity), the FDA, EMA, CDSCO, and WHO have the responsibility of developing technical standards and audit processes, the developers/industry have the responsibility of providing lightweight and adaptive FL stacks as well as fair training, and the hospitals/health systems have the responsibility of implementing device profiling, regional edge hubs, and cognizant resource participation. In the next five years, coordinated progress across these groups will be essential to ensure safe, equitable, and scalable FL deployment in real-world healthcare environments [153–155].

5.2 Data Heterogeneity: Non-IID Data Challenges

In FL, healthcare data comes from different sources, which include hospitals, clinics, and different medical devices. Data from different sources is rarely identical, and this is attributed to differences in populations, disease distributions, and sampling. As a result, the non-IID data imbalance makes each individual model learn different patterns, which can be termed as 'specialized.' However, when individual models are combined, the overall performance of the global model is significantly poor, and it can be considered over-biased or even heterogeneous [156, 157].

In order to solve this problem, various methods, like FedProx, FedAvgM, and Agnostic Federated Learning (robust aggregation), as well as personalized federated learning (local adaptation with shared core knowledge), have been used. Other techniques include the clustering of similar clients and the harmonization of the data, i.e., the alignment of ICD codes, etc. Advanced techniques include the use of meta-learning and multi-task learning, which help in the rapid adaptation of the model to new datasets. Tackling non-IID data is critical to ensure fairness, inclusivity, and reliability of FL-driven healthcare AI [158–160].

In conjunction with the technical solutions outlined in Table 5, an overall action plan is also required, which includes the involvement of various stakeholders, such as the regulatory agencies, developers, and hospitals. In this regard, the regulatory agencies, such as the FDA, EMA, CDSCO, and WHO, need to establish guidelines for the management of non-IID data. For the developers, the development of scalable FL systems with robust and personalized

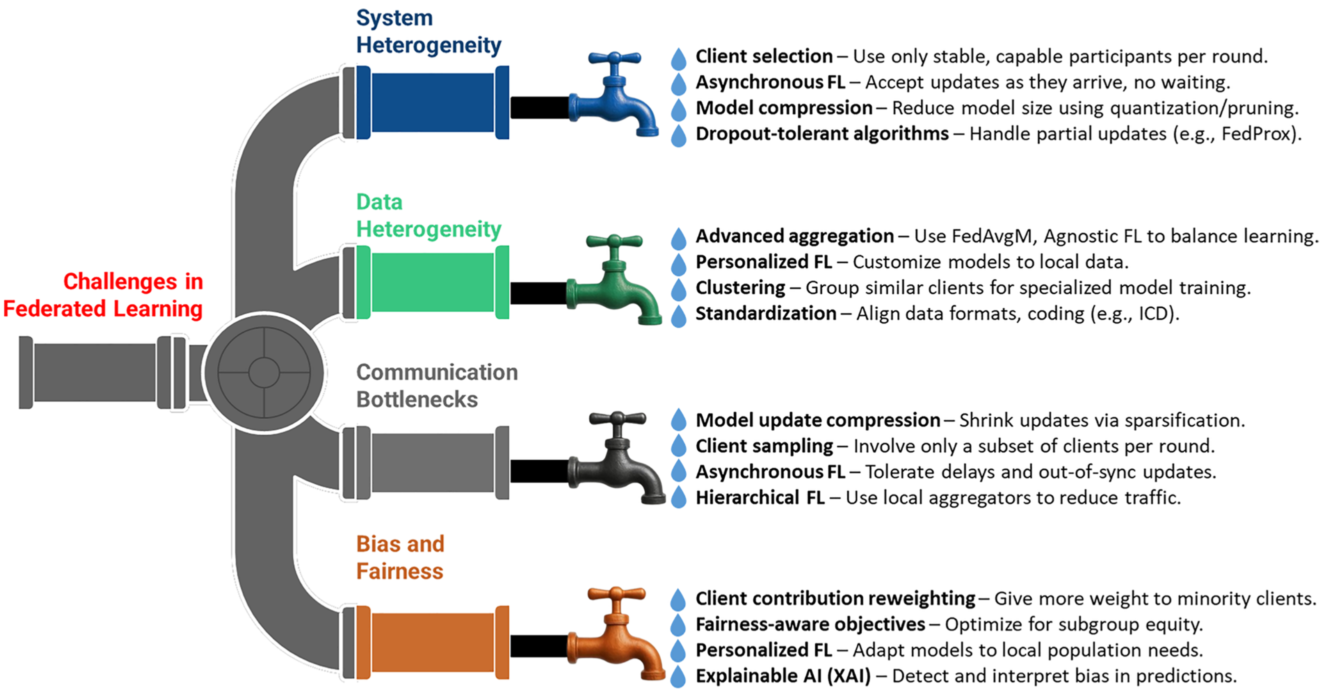


Fig. 10 Visual representation of key challenges in FL and their targeted solution strategies

algorithms is recommended, while the hospitals need to adopt standardized coding, participate in data sharing, and engage in the monitoring of fairness for various patient groups. Together, these steps can help overcome heterogeneity and ensure that FL models serve diverse patient populations equitably [161–163].

5.3 Quantitative Evaluation of Personalization in FL

The personalization effect for FL may be assessed using quantitative metrics by comparing the federated global model to client-specific personalized models in terms of each client's local test data. The personalization effect is expressed in terms of the gap between personalized FL and the baseline approach, such as the FedAvg algorithm. Metrics that can be used in different tasks are the area under curve (AUC), accuracy, F1-score, sensitivity/specificity, Dice score, IoU, root mean square error (RMSE), calibration error, or early warning time. Crucially, besides presenting the mean results, the research should also present measures like the spread of results, the performance at the worst-performing site, and the performance of subgroups of clients, as well as the fraction of clients benefiting from personalization [154].

5.4 Communication Bottlenecks: Bandwidth and Dropout

In FL, communication is key. Clients have to send their updates to the central server over and over again. This might cause a bottleneck. In several rural hospitals or on wearable devices, the bandwidth of the internet may not be large enough, and clients may also experience dropouts due to power outages, depletion of batteries, or unstable connections in general. This makes the training process slow and weakens the global model. Furthermore, clients may also experience several dropouts that cause biases in learning. This has negative implications for the model's performance on those groups [164, 165].

A number of methods are used to solve these issues of communication. Methods such as quantization, sparsification, and delta encoding are used for compression of the models. Sampling of clients is used to reduce the number of clients that are used for sending updates concurrently. Asynchronous FL allows each client to send the updates whenever they are ready without waiting for others. Adaptive communication frequency allows the clients to send updates less frequently, even after a large number of local training iterations. Hierarchical FL inserts regional servers between local clients and the main server, cutting down long-distance communication. These methods together enhance scalability, reliability, and ease of use across diverse healthcare environments for FL [166, 167].

Table 4 Action Plan for Addressing System heterogeneity in Federated learning (Next 5 Years)

Stakeholder	0–12 Months (Now)	12–24 Months	3–5 Years
Regulators (FDA, EMA, CDSCO, WHO)	Draft guidance for FL in healthcare; define system heterogeneity risks; require telemetry for device performance; introduce model cards (hardware/latency/fairness).	Issue technical specs for audits (logs, failures, update frequency); endorse asynchronous FL; launch regulatory sandboxes incl. LMICs.	Global harmonization of conformity frameworks; publish fairness benchmarks; mandate interoperability and edge–cloud safety cases.
Developers & Industry	Client/resource profiling; tiered client selection; asynchronous/partial updates; adaptive compression (8-bit/4-bit); triggers to train on charge+Wi-Fi.	Deploy personalized FL (clustered/proximal); model distillation for wearables; split/edge-cloud training; heterogeneitydashboards.	Standardized FL stacks robust to heterogeneity; energy- and cost-aware training; automated fairness reweighting for low-resource sites.
Hospitals & Health Systems	Inventory devices/connectivity; whitelist stable clients; schedule training off-peak; deploy thin models to weak devices; collect telemetry.	Adopt asynchronous cohorts; SLA-based participation (signal/battery); local fine-tuning via model marketplaces; playbooks for dropouts.	Build regional edge hubs for pre-aggregation; integrate QoS & fairness targets; refresh device fleets; share anonymized logs for benchmarking.

Table 5 Action Plan for Addressing data heterogeneity in Federated learning (Next 5 Years)

Stakeholder	0–12 Months (Now)	12–24 Months	3–5 Years
Regulators (FDA, EMA, CDSCO, WHO)	Publish guidance on handling non-IID data in AI submissions; mandate reporting of data diversity and cohort composition.	Require validation across heterogeneous cohorts; define standards for ICD/coding harmonization; support regulatory sandboxes for FL pilots.	Establish global benchmarks for fairness & inclusivity; require ongoing monitoring of performance drift across subgroups.
Developers & Industry	Integrate robust algorithms (FedProx, Agnostic FL); support local fine-tuning; build tools for data profiling and cohort visualization.	Deploy clustering-based FL and personalized FL frameworks in production; expand libraries for standardized preprocessing pipelines.	Advance meta-learning/multi-task FL stacks; build adaptive global models that self-adjust to incoming client distributions.
Hospitals & Health Systems	Map current data standards; adopt basic harmonization; report cohort descriptors during FL training.	Form data-sharing consortia with aligned preprocessing; cluster participation by cohort similarity; run pilot studies with harmonized EHR + wearable inputs.	Establish regional hubs for harmonized datasets; adopt continuous fairness monitoring; link outcomes to clinical registries.

In other words, to resolve communication bottlenecks effectively, it is necessary to take both technological and governance measures. More concretely, as shown in Table 6 (Action Plan for Addressing Communication Bottlenecks in FL), it is necessary for the regulators to set standards for reporting the utilization of the bandwidth and dropout rates, for the developers to integrate the use of compression and first-hop/adaptive communication into the FL stacks, and for the hospitals to set up regional hubs and report on the reliability of the communication. Coordinated progress across these groups will be essential to make FL truly ready for global, resource-diverse healthcare deployment [168, 169].

5.5 Bias and Fairness: Ethical Concerns

One of the primary concerns in the context of the healthcare FL paradigm is fairness. Although FL has the advantage of aggregating large amounts of data from various sources, it does not ensure the fair representation of the sources in the aggregated model. Most often, the well-connected and resource-rich hospitals have higher participation rates, while the smaller clinics are less represented in the model, which

Table 6 Action Plan for Addressing communication bottlenecks in Federated learning (Next 5 Years)

Stakeholder	0–12 Months (Now)	12–24 Months	3–5 Years
Regulators (FDA, EMA, CDSCO, WHO)	Issue guidance on communication efficiency metrics in FL pilots; require reporting of dropout and bandwidth usage.	Define standards for compressed updates; include robustness-to-dropout in regulatory sandbox trials.	Establish global benchmarks for FL reliability under low-resource networks; mandate energy/bandwidth disclosures in submissions.
Developers & Industry	Implement compression methods (quantization, sparsification, delta encoding); client sampling; asynchronous updates.	Deploy adaptive update frequency; develop hierarchical FL frameworks with edge aggregators; optimize communication stacks.	Advance energy- and cost-aware FL protocols; integrate automated dropout recovery and fairness-aware aggregation.
Hospitals & Health Systems	Monitor local bandwidth/connectivity; whitelist stable devices; schedule training off-peak; invest in reliable Wi-Fi/power backups.	Pilot regional hubs to pre-aggregate updates; track dropout and latency logs; implement adaptive frequency protocols.	Establish network resilience programs; integrate FL into hospital IT infrastructure; share anonymized communication performance logs.

may, in turn, lead to biased results for the majority or for certain age groups, ethnicities, or medical conditions. This raises serious ethical concerns since biased models could worsen health inequity [170, 171].

In response to these risks, various mitigation strategies have been proposed to reduce the impact of the risks on the research findings. For example, reweighting the updates from the clients can enhance the updates from the smaller sites or sites with fewer participants. Fairness-aware training methods, such as group regularization or constraints, have been developed to control the learning process in a fair manner. Personalized FL allows for adaptation without compromising the global knowledge, and the use of XAI helps in the identification and correction of biases in the predictions [153, 172, 173].

The extensive overview provided in Table 7 (Action Plan for Promoting Fairness in FL) reveals that to overcome these issues, several actions are required to be undertaken in coordination with one another. There are needs for regulators to incorporate fairness metrics and sub-group reporting mandates, for developers to design equity-aware aggregation mechanisms and bias detection modules, and for hospitals to provide diverse cohort data and sub-group performance tracking. Thus, fairness in FL is not just a technical requirement but rather a moral responsibility that ensures that healthcare AI systems are not only accurate and private but also fair and equitable to all patients.

Table 7 Action Plan for promoting fairness in Federated learning (Next 5 Years)

Stakeholder	0–12 Months (Now)	12–24 Months	3–5 Years
Regulators (FDA, EMA, CDSCO, WHO)	Issue draft guidance requiring bias reporting in FL models; define minimum fairness metrics (accuracy by subgroup).	Mandate multi-cohort validation before approval; establish fairness audits; integrate equity benchmarks into FL pilots.	Standardize global fairness frameworks; require continuous subgroup monitoring; mandate XAI-based transparency reports.
Developers & Industry	Integrate fairness-aware aggregation (reweighting, proportional sampling); include XAI dashboards to detect bias.	Deploy group-regularized FL and personalized FL frameworks; build toolkits for fairness constraints in production systems.	Advance adaptive equity-aware FL algorithms; integrate automated bias correction modules.
Hospitals & Health Systems	Report demographic and cohort diversity in FL participation; track subgroup performance locally.	Form inclusive FL consortia to balance urban vs. rural contributions; adopt fairness-aware protocols.	Establish regional fairness hubs with representative cohorts; link fairness monitoring to clinical quality metrics.

5.6 Practical Translational Challenges

Most of the existing FL approaches in biomedical and chemical imaging encounter some practical obstacles in real-world deployment. As multiple centers collaborate, they have to handle the misaligned imaging protocols and scanner configurations. Disparities in the acquisition parameters, contrast agent, and local pre-processing pipeline cause domain shifts that can erode the performance of the federated model. The new solutions have focused on establishing consensus protocols, performing site-level quality control procedures, and implementing harmonization methods based on statistics or deep learning that map site-specific data into a more standard representation [174].

Chemical imaging datasets will add yet another layer to this heterogeneity, due in part to variations in spatial and spectral resolutions, calibration procedures, and noise characteristics. This makes it difficult to define a universally applicable single pre-processing pipeline. In this setting, approaches like representation learning, domain adaptation, and model personalization within federated frameworks are particularly well-placed to learn robust shared features while still accounting for the idiosyncrasies of individual sites [175, 176].

Finally, most of the existing work is based on retrospective or sandbox-style evaluation. Translating the federated systems into routine practice will require prospective validation embedded within real clinical workflows-including picture archiving and communication systems, laboratory information systems/electronic health records, and laboratory pipelines-so ongoing performance monitoring, periodic updating of models, and governance structures guaranteeing safety, accountability, and regulatory compliance will be imperative [177, 178].

5.7 Privacy Limitations and Leakage Risks in FL

Although FL is often described as privacy-preserving because raw patient data remain within local institutions, this should not be interpreted as an absolute guarantee of confidentiality. Indeed, model parameters, gradients, or intermediate updates could potentially reveal some confidential information about the training samples of clients, especially in cases of smaller client sample sizes, very rare medical conditions, very imbalanced cohorts, or multiple iterations of the same client. Such attacks include gradient inversion, model inversion, membership inference, and property inference to name a few. Hence, FL is better seen as a privacy-preserving technique rather than a guarantee of privacy [179–181].

The implementation of such a privacy-preserving technique as FL in healthcare settings implies the application of additional technical means to further ensure patient confidentiality. One may use, for instance, secure aggregation of model updates to protect them from analysis by the server. Differential privacy may be applied to constrain the impact of each individual or institution on model updates. One could also employ homomorphic encryption or secure multiparty computation to perform calculations based on encrypted updates and trusted execution environments to protect aggregation runtime processes. Additional techniques may include gradient clipping, update compression, anomaly detection, access control, logging, and budget privacy tracking.

6 Conclusion and Future Prospects

FL is rapidly emerging as a revolutionary technology in healthcare AI, empowering institutions to collaboratively train powerful models without the need for actually sharing sensitive patient data. Here we have highlighted how FL addresses important challenges in data privacy, regulatory compliance, and data decentralization through several real-world applications: medical imaging, EHR, wearable biosensors, and CDSS. FL enables innovation on health AI without compromising trust or legal boundaries by keeping data on premise and sharing updates to the model only. Yet, challenges like system and data heterogeneity, scarce communication bandwidth, and fairness & performance considerations for various populations will impede universal FL adoption. The upcoming wave of FL systems will witness trends like integrating with LLMs & multimodal data where AI learns from rich data sources - complex mixtures of text, images, signals, etc.- forming a more cognitively-complete AI. The development of explainable FL will further cultivate clinical trust by making model predictions interpretable and transparent. Regulatory alignment and the development of standardised frameworks will be imperative for real-world deployment as FL systems seek to comply with evolving global data protection regulations. On the other hand, the advent of self-adaptive FL solutions, capable of automating the adaptation process to device capabilities as well as the variation of data across devices as well as user needs is the continuing evolution of our models. FL is more than a method, it is a shift in thinking of how to connect the technical AI advancement with the ethics and practical requirements of modern health care that are the real constraints that AI has to operate within. Lighting the path between privacy and intelligence, FL gives us a model that enables AI systems to be powerful, precise, trusted, inclusive and deployable across the world.

Vocabulary

AI	AI is a field of study that specializes in developing computational techniques that allow machines to perform tasks that would usually be characterized by human intelligence.

FL	FL is a distributed ML approach in which several data-possessing sites collaborate in training a common model. Importantly, the collaboration happens without raw data being shared; rather, model parameters or updates are transmitted to a central server where they are aggregated to refine the shared model.
Chemical imaging	It describes a suite of techniques that includes Raman spectroscopy, infrared imaging, and mass spectrometry imaging-all of which are capable of rendering maps reflecting the chemical composition of a sample with spatial resolution
Wearable sensors	This are portable, miniaturized sensing devices that can be embedded into common items worn as clothes, accessories, or on-skin platforms.
Biosensors	A biosensor is an analytical device that couples a biological recognition element, like enzymes, antibodies, or receptors, with a physical transducer. This combination converts a biochemical interaction into a quantifiable signal.

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Competing Interests The authors declare no competing interests.

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